Odd Perfect Numbers as Edge-Weighted Digraphs

Notes on a σ-graph program, a stronger graph-theoretic approach, and a Poisson-equation reformulation

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Abstract

These notes record a discussion of an LLM-generated research program (Gemini/Argon and ChatGPT/Sol and Claude/Opus) that attacks the odd perfect number (OPN) problem through a weighted directed graph on the prime divisors of a hypothetical OPN, the *σ-graph*. The program builds on a 2026 preprint of Rodgers, whose basic order criterion for edges (Theorem 5.1 there) is false; the corrected criterion is given below. Sections 1 and 2 set up the general number theory and graph constructions for an arbitrary integer. Section 3 specializes them to odd perfect numbers. Section 4 summarizes the proposed lemmas, with examples. Section 5 assesses them: the arithmetic is correct, the small lemmas are mostly classical facts in graph language, and the two "breakthrough lemmas" are reformulations of the conjecture rather than reductions of it. Section 6 sketches a more realistic graph-theoretic program built on discharging, linear-programming duality, and several exact or partial undirected reformulations of the σ-graph. Along the way we prove a few small results of our own: the σ-graph of an OPN is strongly connected (so its Laplacian has rank $\omega(N) - 1$), most of its logarithmic flow travels around cycles, and no odd perfect or multiperfect number can have a *symmetric* (undirected) σ-graph—for OPNs this is just the handshake lemma. Section 7 summarizes the discussion around a single observation: an OPN is equivalent to a Poisson equation on its σ-graph, with the log-prime vector as potential. It then asks what harmonic morphisms and equitable partitions can and cannot do with that equation. All numerical examples were checked by computer. Companion Python code, with a test suite reproducing every numerical example, is at `https://github.com/wdjoyner/opn-sigma-graphs`.

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How to read these notes

The central idea is a dictionary between arithmetic and graph theory. A prime divisor of a hypothetical OPN becomes a vertex; a prime appearing in a local divisor sum becomes a directed edge; its r-adic valuation becomes the edge weight; and perfectness fixes the total incoming weight at each vertex. Cyclotomic factorization then restricts which edges are allowed. The table below is a useful guide to the rest of the paper.

Number-theoretic object	**Digraph interpretation**
prime $q \mid N$	vertex q
$q^a \parallel N$	vertex q carrying exponent/demand a
$r \mid \sigma(q^a)$	directed edge $q \to r$
$v_r(\sigma(q^a))$	weight of $q \to r$
$\sum_q v_r(\sigma(q^{a_q})) = a_r$	exact incoming-weight balance at r
$\Phi_d(q) \mid \sigma(q^a)$	cyclotomic layer or "obligation" at q
primitive divisor of $q^d - 1$	forced order-d target
$\prod_q \Sigma_q = N$	global multiplicative closure
Euler prime p	unique sink for logarithmic divergence

A reader mainly interested in OPN arithmetic can regard the graph language as bookkeeping for simultaneous σ-chains. A reader coming from graph theory should keep in mind that the difficult information is not generic connectivity or spectrum: it is the arithmetic restriction on which integer weights can occur. Sections 4 and 5 deliberately separate these two issues.

The notes are organized from the general to the special.

- Section 1 collects the elementary number theory: divisor functions, abundancy, valuations, orders and cyclotomic polynomials.
- Section 2 builds the graphs for an *arbitrary* integer: the σ-graph, its Poisson equation, harmonic morphisms, arithmetical structures, undirected encodings, and divisor graphs.
- Section 3 specializes everything to a hypothetical odd perfect number.
- Sections 4–7 discuss the AI-generated program, assess it, and develop a more realistic approach.

A glossary of technical terms is in Appendix A.

1 Elementary number theory

This section collects the number theory used later. Everything here holds for all positive integers. Odd perfect numbers appear only from Section 3 on.

1.1 Divisor functions

Definition 1.1 (Divisor functions)**.** For a real number r and a positive integer n, put

$$\sigma_r(n) = \sum_{d \mid n} d^r.$$

Two cases matter most. The function σ_0 counts divisors, because each divisor contributes $d^0 = 1$. The function σ_1 adds the divisors up. Both are often given other names: σ_0 is written τ or d, and σ_1 is written σ. These notes write σ_0 for the divisor-counting function and, following custom, σ for σ_1.

Example 1.2 ($n = 12$)**.** The divisors of 12 are $1, 2, 3, 4, 6, 12$. So

$$\sigma_0(12) = 6, \qquad \sigma(12) = 28, \qquad \sigma_2(12) = 1 + 4 + 9 + 16 + 36 + 144 = 210, \qquad \sigma_{-1}(12) = \tfrac{28}{12} = \tfrac{7}{3}.$$

Proposition 1.3 (Multiplicativity)**.** *If* $\gcd(m, n) = 1$*, then* $\sigma_r(mn) = \sigma_r(m)\,\sigma_r(n)$ *for every* r*.*

Proof. Every divisor of mn factors uniquely as de with $d \mid m$ and $e \mid n$. Hence $\sum_{f \mid mn} f^r = \sum_{d \mid m} \sum_{e \mid n} d^r e^r = \sigma_r(m)\sigma_r(n)$. □

Proposition 1.4 (Prime powers)**.** *Let* p *be prime and* $a \geq 0$*. Then*

$$\sigma_r(p^a) = 1 + p^r + p^{2r} + \cdots + p^{ar} = \begin{cases} \dfrac{p^{r(a+1)} - 1}{p^r - 1}, & r \neq 0, \\ a + 1, & r = 0. \end{cases}$$

Proof. The divisors of p^a are $1, p, \ldots, p^a$. Sum their r-th powers; for $r \neq 0$ this is a geometric series. □

Corollary 1.5. *If* $n = \prod_i p_i^{a_i}$*, then*

$$\sigma_0(n) = \prod_i (a_i + 1), \qquad \sigma(n) = \prod_i \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

For $12 = 2^2 \cdot 3$ this gives $\sigma_0(12) = 3 \cdot 2 = 6$, $\sigma(12) = 7 \cdot 4 = 28$ and $\sigma_2(12) = 21 \cdot 10 = 210$, as in Example 1.2.

Proposition 1.6 (Reflection)**.** *For every* r*,* $\sigma_{-r}(n) = n^{-r}\sigma_r(n)$*. In particular* $\sigma_{-1}(n) = \sigma(n)/n$*.*

Proof. The map $d \mapsto n/d$ permutes the divisors of n. So $\sum_{d \mid n} d^{-r} = \sum_{d \mid n} (n/d)^{-r} = n^{-r} \sum_{d \mid n} d^r$. □

Definition 1.7 (Local divisor sums)**.** If p^a is the exact power of the prime p dividing n, the number $\sigma(p^a) = 1 + p + \cdots + p^a$ is the *local divisor sum* of n at p.

By Proposition 1.3, $\sigma(n)$ is the product of its local divisor sums. For 12 these are $\sigma(4) = 7$ and $\sigma(3) = 4$, and $7 \cdot 4 = 28$. This one fact drives the whole graph construction of Section 2.

1.2 Abundancy; perfect and multiperfect numbers

Definition 1.8 (Abundancy)**.** The *abundancy* of n is $h(n) = \sigma(n)/n = \sigma_{-1}(n)$.

- n is *perfect* if $h(n) = 2$, i.e. n is the sum of its proper divisors.
- n is *abundant* if $h(n) > 2$ and *deficient* if $h(n) < 2$.
- n is *k-perfect*, or *multiperfect*, if $h(n) = k$ is an integer $k \geq 2$.

Lemma 1.9 (Basic facts about abundancy)**.** *(a) h is multiplicative: $h(mn) = h(m)h(n)$ when $\gcd(m, n) = 1$.*

(b) For a prime power, $1 < h(p^a) = 1 + \frac{1}{p} + \cdots + \frac{1}{p^a} < \frac{p}{p-1}$.

(c) If $d \mid n$ and $d < n$, then $h(d) < h(n)$.

Proof. (a) follows from Proposition 1.3. (b) is a partial sum of the geometric series $\sum_{j \geq 0} p^{-j} = p/(p-1)$. (c) Every divisor of d divides n, and n itself does not divide d, so $h(d) = \sum_{e \mid d} 1/e < \sum_{e \mid n} 1/e = h(n)$ by Proposition 1.6. □

Example 1.10 (A small catalog)**.**

n	factorization	$h(n)$	type
6	$2 \cdot 3$	2	perfect
28	$2^2 \cdot 7$	2	perfect
12	$2^2 \cdot 3$	7/3	abundant
15	$3 \cdot 5$	8/5	deficient
45	$3^2 \cdot 5$	26/15	deficient
945	$3^3 \cdot 5 \cdot 7$	128/63	abundant (smallest odd abundant number)
120	$2^3 \cdot 3 \cdot 5$	3	3-perfect

Theorem 1.11 (Euclid–Euler)**.** *An even number is perfect if and only if it equals $2^{p-1}M_p$, where $M_p = 2^p - 1$ is prime (a* Mersenne prime*).*

The "if" direction is a one-line check: $\sigma(2^{p-1}M_p) = (2^p - 1)(M_p + 1) = M_p \cdot 2^p = 2 \cdot 2^{p-1}M_p$. For $p = 2$ and $p = 3$ this gives 6 and 28. No odd perfect number is known. The *odd perfect number (OPN) conjecture* says that none exists.

Definition 1.12 (Spoofs)**.** A *spoof odd perfect number* is a formal perfect factorization in which a composite odd factor is treated as though it were prime when the local divisor sum is computed. In the common one-spoof-factor form, an odd $n = km$ with $\gcd(k, m) = 1$ satisfies

$$\sigma(k)\,(m + 1) = 2km,$$

with the composite m playing the role of a *spoof prime*. This does not make n perfect, because $\sigma(m) \neq m + 1$ for composite m.

Example 1.13 (The Descartes spoof)**.** Let $D = 3^2 \cdot 7^2 \cdot 11^2 \cdot 13^2 \cdot 22021 = 198585576189$. Pretend that 22021 is prime. Then

$$\sigma(3^2) = 13, \quad \sigma(7^2) = 3 \cdot 19, \quad \sigma(11^2) = 7 \cdot 19, \quad \sigma(13^2) = 3 \cdot 61, \quad 22021 + 1 = 2 \cdot 7 \cdot 11^2 \cdot 13,$$

and the product is exactly $2D$. The trick is that $22021 = 19^2 \cdot 61$ is assembled from the 19's supplied by 7^2 and 11^2 and the 61 supplied by 13^2. Spoofs are studied systematically by Dittmer [13] and the BYU Computational Number Theory Group [9].

1.3 Prime factorizations

Definition 1.14. Let r be a prime.

- For a nonzero integer x, the *r-adic valuation* $v_r(x)$ is the exponent of r in x. It extends to nonzero rationals by $v_r(x/y) = v_r(x) - v_r(y)$.
- $\omega(n)$ is the number of distinct primes dividing n, and $\Omega(n) = \sum_r v_r(n)$ counts them with multiplicity.
- We write $r^a \parallel n$ when $v_r(n) = a$. The prime power r^a is then a *component* of n.
- A divisor $M \mid n$ is *unitary* if $\gcd(M, n/M) = 1$, i.e. M is a product of some of the components of n.
- $\varphi(n)$, Euler's totient, counts $1 \le j \le n$ with $\gcd(j, n) = 1$.

Example 1.15. For $n = 360 = 2^3 \cdot 3^2 \cdot 5$: $v_2(360) = 3$, $\omega(360) = 3$ and $\Omega(360) = 6$. The unitary divisors of $12 = 2^2 \cdot 3$ are $1, 3, 4, 12$; the divisors 2 and 6 are not unitary, since $\gcd(2, 6) = 2$.

The valuation turns products into sums: $v_r(xy) = v_r(x) + v_r(y)$. Unique factorization says that a positive rational is determined by its valuations. By Lemma 1.9(a), $h(n) = h(M)\,h(n/M)$ for every unitary divisor M.

1.4 Multiplicative order and cyclotomic polynomials

Definition 1.16 (Multiplicative order)**.** Let r be prime and $r \nmid s$. The *order* $\operatorname{ord}_r(s)$ is the least $d \ge 1$ with $s^d \equiv 1 \pmod r$.

Lemma 1.17. *Let $d = \operatorname{ord}_r(s)$. Then $d \mid r - 1$, and $r \mid s^n - 1$ if and only if $d \mid n$.*

Proof. The residues of $1, s, s^2, \ldots$ repeat with period d. So $s^n \equiv 1$ exactly when $d \mid n$. Fermat's little theorem gives $s^{r-1} \equiv 1$, hence $d \mid r - 1$. □

Example 1.18. $13 \equiv 1 \pmod 3$, so $\operatorname{ord}_3(13) = 1$. Also $13^3 \equiv 1 \pmod{61}$ and $61^3 \equiv 1 \pmod{13}$, while neither base is $\equiv 1$; so $\operatorname{ord}_{61}(13) = \operatorname{ord}_{13}(61) = 3$. Finally $\operatorname{ord}_{11}(3) = 5$, since $3^5 = 243 = 22 \cdot 11 + 1$.

Definition 1.19 (Cyclotomic polynomials)**.** The d-th *cyclotomic polynomial* $\Phi_d(x)$ is the monic polynomial whose roots are the primitive d-th roots of unity. It has integer coefficients, degree $\varphi(d)$, and satisfies $x^n - 1 = \prod_{d \mid n} \Phi_d(x)$.

For example $\Phi_1(x) = x - 1$, $\Phi_3(x) = x^2 + x + 1$ and $\Phi_9(x) = x^6 + x^3 + 1$. Dividing $x^{a+1} - 1$ by $\Phi_1(x) = x - 1$ gives the factorization that matters most here.

Proposition 1.20 (Cyclotomic layers)**.** *For every integer $s \ge 2$ and $a \ge 1$,*

$$\sigma(s^a) = \frac{s^{a+1} - 1}{s - 1} = \prod_{\substack{d \mid a+1 \\ d > 1}} \Phi_d(s).$$

Each factor $\Phi_d(s)$ is a *cyclotomic layer* of $\sigma(s^a)$. For example $\sigma(3^8) = \Phi_3(3)\,\Phi_9(3) = 13 \cdot 757 = 9841$, with two layers because 9 has two divisors larger than 1. A prime exponent $a + 1$ gives a single layer: $\sigma(3^{12}) = \Phi_{13}(3) = 797161$, which is itself prime.

Definition 1.21 (Primitive prime divisor)**.** A prime ℓ is a *primitive prime divisor* of $s^d - 1$ if $\ell \mid s^d - 1$ but $\ell \nmid s^j - 1$ for $1 \le j < d$. Equivalently, $\mathrm{ord}_\ell(s) = d$.

By Lemma 1.17, a primitive prime divisor ℓ of $s^d - 1$ satisfies $\ell \equiv 1 \pmod{d}$. "Primitive" refers to the fixed base s and exponent d. It does not mean that ℓ is new in any other sense.

1.5 Two standard tools

Theorem 1.22 (Lifting the exponent, LTE)**.** *Let r be an odd prime and x an integer with $r \mid x-1$. Then for every $n \ge 1$, $v_r(x^n - 1) = v_r(x - 1) + v_r(n)$.*

Theorem 1.23 (Bang–Zsigmondy)**.** *Let $s > 1$ and $d > 1$. Then $s^d - 1$ has a primitive prime divisor, except when $(s, d) = (2, 6)$, or $d = 2$ and $s + 1$ is a power of 2.*

For odd s and odd $d > 1$ neither exception applies. A primitive prime ℓ of $s^d - 1$ then satisfies $d \mid \ell - 1$; since ℓ and d are both odd, $\ell \equiv 1 \pmod{2d}$, and in particular $\ell \ge 2d + 1$. Moreover $\ell \nmid s - 1$, so ℓ divides the layer $\Phi_d(s)$ and hence $\ell \mid \sigma(s^a)$ whenever $d \mid a + 1$.

Example 1.24. For $s = 13$ and $d = 3$: $13^3 - 1 = 2^2 \cdot 3^2 \cdot 61$. The primes 2 and 3 divide $13 - 1$, so they are not primitive. The prime 61 is primitive, and indeed $61 \equiv 1 \pmod{6}$.

1.6 Valuations of local divisor sums

The next result computes the exponent of a prime r in a local divisor sum. It is the arithmetic heart of the σ-graph.

Proposition 1.25 (Order/LTE classification)**.** *Let $s \ne r$ be odd primes, let $a \ge 1$, and put $n = a+1$ and $d = \mathrm{ord}_r(s)$. Then $v_r(\sigma(s^a)) = v_r\big((s^n - 1)/(s - 1)\big)$, and:*

I. if $d \nmid n$, then $v_r(\sigma(s^a)) = 0$;

II. if $d = 1$ (i.e. $r \mid s - 1$), then $v_r(\sigma(s^a)) = v_r(n)$; *(L2a)*

III. if $d > 1$ and $d \mid n$, then $v_r(\sigma(s^a)) = v_r(s^d - 1) + v_r(n/d)$. *(L2b)*

Proof. Case I: $r \mid s^n - 1$ iff $d \mid n$ (Lemma 1.17), and if $d > 1$ then $r \nmid s - 1$. Case II: LTE gives $v_r(s^n - 1) = v_r(s - 1) + v_r(n)$; subtract $v_r(s - 1)$. Case III: $r \nmid s - 1$, and LTE applied to $x = s^d$ gives $v_r(s^n - 1) = v_r(s^d - 1) + v_r(n/d)$. □

So valuation can come from three sources: an *order-one* exponent factor $v_r(n)$; *Wieferich-type* lifting $v_r(s^d - 1) > 1$; and *multiplier* lifting $v_r(n/d) > 0$.[1]

Example 1.26 (All three sources)**.** (a) *Order one.* For $\sigma(13^2) = 183 = 3 \cdot 61$: $n = 3$ and $\mathrm{ord}_3(13) = 1$, so $v_3(\sigma(13^2)) = v_3(3) = 1$.

(b) *Wieferich-type.* $\sigma(3^4) = 121 = 11^2$. Here $\mathrm{ord}_{11}(3) = 5$ and $3^5 - 1 = 242 = 2 \cdot 11^2$, so $v_{11}(\sigma(3^4)) = v_{11}(3^5 - 1) = 2$. (Incidentally $(3^5 - 1)/2 = 11^2$ is one of the known solutions of the Nagell–Ljunggren equation $\frac{x^n-1}{x-1} = y^m$.)

[1] Rodgers' preprint (Remark 5.1, §10.1, and Appendix D.3) asserts that the last two are "the only sources of valuation amplification" on edges from nonspecial primes. The order-one channel gives a counterexample: $\sigma(7^8) = 3^2 \cdot 19 \cdot 37 \cdot 1063$, and since $\mathrm{ord}_3(7) = 1$ and $n = 9$, Case II gives $v_3(\sigma(7^8)) = v_3(9) = 2$. This valuation is 2, but it is not covered by the preprint's two-term decomposition $v_r(q^d - 1) + v_r(e/d)$ (its Theorem 5.2), which assumes $d > 1$. Applying that formula anyway with $d = 1$ would give $v_3(7 - 1) + v_3(9) = 3$, which is wrong, because the factor $v_3(7 - 1) = 1$ is cancelled by the denominator $q - 1$ of $\sigma(q^8) = (q^9 - 1)/(q - 1)$.

(c) *Multiplier.* $\mathrm{ord}_{13}(3) = 3$ and $3^3 - 1 = 26$. With $n = 39$, Case III gives $v_{13}(\sigma(3^{38})) = v_{13}(26) + v_{13}(13) = 2$, which can be confirmed by direct computation.

Remark 1.27 (Which layer carries which valuation). Proposition 1.20 writes $\sigma(s^a)$ as a product of layers $\Phi_m(s)$, $m \mid n$, $m > 1$. By the classical theorem of Sylvester and Lucas on prime divisors of cyclotomic values, with $d = \mathrm{ord}_r(s)$:

- If $d > 1$ and $d \mid n$, then $v_r(\Phi_d(s)) = v_r(s^d - 1) \geq 1$ carries the base and Wieferich-type valuation, while each layer $\Phi_{dr^k}(s)$ with $dr^k \mid n$ and $k \geq 1$ contributes exactly one factor of r.
- If $d = 1$, then $v_r(\Phi_{r^k}(s)) = 1$ for each $1 \leq k \leq v_r(n)$.
- All other layers are prime to r.

This layer-by-layer bookkeeping is used for the cyclotomic blow-up in Section 7.4.

2 Graph constructions

This section sets up the graphs. Again everything is general: the constructions apply to every integer $n > 1$. The examples use the numbers of Example 1.10.

2.1 Weighted digraphs and their Laplacians

Definition 2.1 (Weighted digraphs)**.** A *weighted digraph* on a finite set V is given by a matrix $W = (w(s, r))_{s,r \in V}$ of nonnegative integers. Rows index sources and columns index targets. There is an edge $s \to r$ exactly when $w(s, r) > 0$.

- The *in-degree* of r is $\sum_s w(s, r)$, a column sum.
- The *out-degree* of s is $\sum_r w(s, r)$, a row sum.
- The digraph is *strongly connected* if every vertex can reach every other vertex along directed edges.

Two facts about strong connectivity are used repeatedly. First, a digraph is strongly connected if and only if every nonempty proper set $A \subset V$ has an edge entering it. (If not, take A to be the set of vertices that cannot be reached from some fixed vertex.) Second, collapsing each strongly connected component to a point gives an acyclic *condensation*, which always has at least one *sink* component, with no edges leaving it.

Definition 2.2 (Laplacians of a digraph)**.** For a diagonal matrix D, put $L = D - W$.

- If D holds the in-degrees, L is the *in-degree Laplacian*. Its columns sum to 0: $\mathbf{1}^{\mathsf{T}} L = 0$.
- If D holds the out-degrees, L is the *out-degree Laplacian*. Its rows sum to 0: $L\mathbf{1} = 0$.

The in-degree Laplacian conserves total mass when it acts on row vectors from the left; the out-degree Laplacian kills constants when it acts on column vectors from the right. Both conventions occur in the literature on directed graphs (Chung [12], Butler–Chung [8], Holroyd et al. [15]). The arithmetic Laplacian of these notes is of in-degree type, but it acts on a potential from the right.

Theorem 2.3 (Perron–Frobenius)**.** *Let M be a nonnegative square matrix whose support digraph is strongly connected. Then the spectral radius $\rho(M)$ is a simple eigenvalue of M with an eigenvector whose entries are all positive.*

For undirected graphs this is the familiar statement about the largest adjacency eigenvalue; see the subsection "Perron–Frobenius" of the chapter "Applications to graph spectra" of [18].

Example 2.4 (A 2-cycle)**.** Take $V = \{2, 3\}$ with $w(2, 3) = 1$ and $w(3, 2) = 2$. The in-degrees are $(2, 1)$ and the out-degrees are $(1, 2)$. The in-degree Laplacian is $L = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}$; its columns sum to 0, its rows do not. This digraph is the σ-graph of 6 (Example 2.7).

2.2 The σ-graph of an integer

Definition 2.5 (σ-graphs of an integer)**.** Let $n > 1$. Let $\widehat{S}(n)$ be the set of primes dividing $n\,\sigma(n)$, and put $a_r = v_r(n)$ for $r \in \widehat{S}(n)$; thus $a_r = 0$ when $r \nmid n$. The *extended σ-graph* $\widehat{G}_\sigma(n)$ is the weighted digraph on $\widehat{S}(n)$ with

$$w(s, r) = v_r\big(\sigma(s^{a_s})\big).$$

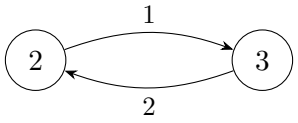
A vertex with $a_s = 0$ has $\sigma(s^0) = 1$ and no outgoing edges. Call it a *boundary vertex*. The *internal σ-graph* $G_\sigma(n)$ is the induced subgraph on the primes dividing n.

An edge $s \to r$ says that the prime r occurs in the local divisor sum at s. Its weight says how often. There are no loops, since $\sigma(r^a) \equiv 1 \pmod r$. Every prime of every local divisor sum is a vertex of the extended graph, so the out-degree of s is $\Omega(\sigma(s^{a_s}))$. Proposition 1.25 computes every weight between odd primes.

Definition 2.6 (σ-chains)**.** A *σ-chain* (or factor chain) is a sequence of primes $q_0, q_1, q_2, \ldots$ with $q_{i+1} \mid \sigma(q_i^{a_{q_i}})$ at every step. It is exactly a directed walk in the σ-graph.

Factor chains are a classical tool in computational work on odd perfect numbers. Brent and Cohen [4] introduced them as an algorithm for proving lower bounds, and Brent–Cohen–te Riele [3] organized the case analysis as a branching *proof tree* to show $N > 10^{300}$. The weighted σ-graph packages all chains at once and keeps the multiplicities that a chain forgets. Bibby–Vyncke–Zelinsky [7] use a weighted digraph of the same kind, with the arrows reversed: they draw $p_i \to p_j$ when $p_i \mid \sigma(p_j^{a_j})$. In either convention their $\sigma_{m,n}$-pairs are directed 2-cycles.

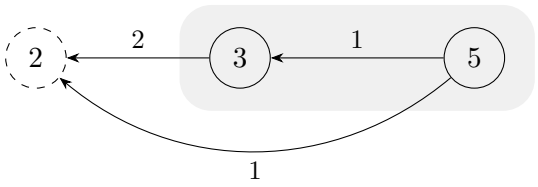
Example 2.7 (The perfect numbers 6 and 28)**.** For $6 = 2 \cdot 3$: $\sigma(2) = 3$ and $\sigma(3) = 4 = 2^2$. So $\widehat{S}(6) = \{2, 3\}$, and the internal and extended graphs coincide:



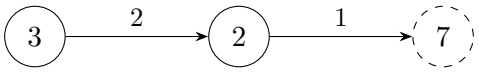
For $28 = 2^2 \cdot 7$: $\sigma(4) = 7$ and $\sigma(7) = 2^3$, giving the edges $2 \xrightarrow{1} 7$ and $7 \xrightarrow{3} 2$.

Example 2.8 (Even perfect numbers)**.** For $N = 2^{p-1}M_p$ the local divisor sums are $\sigma(2^{p-1}) = M_p$ and $\sigma(M_p) = 2^p$. The σ-graph is the 2-cycle $2 \xrightarrow{1} M_p \xrightarrow{p} 2$. The in-degrees are p at 2 and 1 at M_p.

Example 2.9 (A deficient number: 15)**.** For $15 = 3 \cdot 5$: $\sigma(3) = 2^2$ and $\sigma(5) = 2 \cdot 3$. The extended graph on $\{2, 3, 5\}$ has edges $3 \xrightarrow{2} 2$, $5 \xrightarrow{1} 2$ and $5 \xrightarrow{1} 3$. The vertex 2 is a boundary vertex. The internal graph on $\{3, 5\}$ keeps only $5 \to 3$, so it is not strongly connected. In the figure the boundary vertex is drawn dashed, and the internal graph is shaded.



Example 2.10 (An abundant number: 12)**.** For $12 = 2^2 \cdot 3$: $\sigma(4) = 7$ and $\sigma(3) = 2^2$. The extended graph on $\{2, 3, 7\}$ has edges $2 \xrightarrow{1} 7$ and $3 \xrightarrow{2} 2$, and 7 is a boundary vertex:

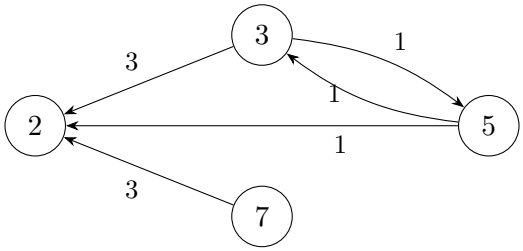


Nothing returns to 3, which is why $d_3(12) = -1$ in Example 2.14.

Example 2.11 (The smallest odd abundant number: 945)**.** For $945 = 3^3 \cdot 5 \cdot 7$,

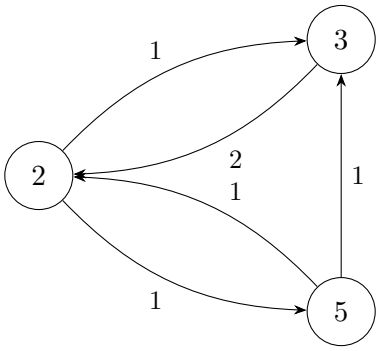
$$\sigma(3^3) = 40 = 2^3 \cdot 5, \qquad \sigma(5) = 6 = 2 \cdot 3, \qquad \sigma(7) = 8 = 2^3.$$

The extended graph on $\{2, 3, 5, 7\}$ has edges $3 \xrightarrow{3} 2$, $3 \xrightarrow{1} 5$, $5 \xrightarrow{1} 2$, $5 \xrightarrow{1} 3$ and $7 \xrightarrow{3} 2$:



Example 2.12 (A 3-perfect number: 120)**.** For $120 = 2^3 \cdot 3 \cdot 5$, $\sigma(120) = 360 = 3 \cdot 120$, and

$$\sigma(2^3) = 15 = 3 \cdot 5, \qquad \sigma(3) = 4 = 2^2, \qquad \sigma(5) = 6 = 2 \cdot 3.$$



In-degrees: 2 receives $2 + 1 = 3$, 3 receives $1 + 1 = 2$, and 5 receives 1. Here every prime of $\sigma(120)$ divides 120, so there are no boundary vertices.

2.3 The defect vector and the Poisson equation

Write $b_r = \sum_s w(s, r)$ for the in-degree of r in $\widehat{G}_\sigma(n)$. Since $\sigma(n)$ is the product of the local divisor sums,

$$b_r = v_r\big(\sigma(n)\big) \qquad (r \in \widehat{S}(n)).$$

Comparing with $a_r = v_r(n)$ gives the basic identity.

Proposition 2.13 (Valuation defect)**.** *Define the* defect vector $d(n) = (b_r - a_r)_{r \in \widehat{S}(n)}$. *Then*

$$h(n) = \frac{\sigma(n)}{n} = \prod_{r \in \widehat{S}(n)} r^{\,d_r(n)}.$$

Consequently n is k-perfect if and only if $d_r(n) = v_r(k)$ for every r. In particular n is multiperfect exactly when $d(n) \geq 0$, and perfect exactly when $d(n)$ is the unit vector e_2 at the prime 2.

Proof. $h(n) = \prod_r r^{v_r(\sigma(n)) - v_r(n)} = \prod_r r^{b_r - a_r}$. A positive rational is determined by its valuations. □

Example 2.14 (Defect vectors)**.**

n	vertices	in-degrees b	exponents a	defect $d(n)$
6	$2, 3$	$(2, 1)$	$(1, 1)$	$(1, 0) = e_2$
28	$2, 7$	$(3, 1)$	$(2, 1)$	$(1, 0) = e_2$
15	$2, 3, 5$	$(3, 1, 0)$	$(0, 1, 1)$	$(3, 0, -1)$
12	$2, 3, 7$	$(2, 0, 1)$	$(2, 1, 0)$	$(0, -1, 1)$
945	$2, 3, 5, 7$	$(7, 1, 1, 0)$	$(0, 3, 1, 1)$	$(7, -2, 0, -1)$
120	$2, 3, 5$	$(3, 2, 1)$	$(3, 1, 1)$	$(0, 1, 0) = e_3$

Each row reproduces $h(n)$. For instance $h(15) = 2^3 \cdot 5^{-1} = 8/5$, and $h(945) = 2^7 \cdot 3^{-2} \cdot 7^{-1} = 128/63$. The odd abundant number 945 overshoots 2, but its defect is far from e_2: the boundary vertex 2 receives weight 7, because all three local divisor sums are even.

The defect has a linear-algebra meaning. Let W be the weight matrix of $\widehat{G}_\sigma(n)$, put $D = \operatorname{diag}(a_s)$ and $L = D - W$. Let $\ell = (\log s)_s$ be the *log-prime vector*, and let $\eta_s = \log h(s^{a_s})$ be the logarithmic local abundancy (so $\eta_s = 0$ at a boundary vertex).

Proposition 2.15 (The Poisson equation of an integer)**.** *For every integer $n > 1$,*

$$L\ell = -\eta, \qquad \mathbf{1}^{\mathsf{T}} L = -d(n)^{\mathsf{T}}, \qquad \sum_s \eta_s = \log h(n).$$

Proof. Row s of $L\ell$ is $a_s \log s - \sum_r w(s, r) \log r = a_s \log s - \log \sigma(s^{a_s}) = -\eta_s$. Column r of L sums to $a_r - b_r = -d_r(n)$. Finally $\sum_s \eta_s = \log \prod_s h(s^{a_s}) = \log h(n)$ by Lemma 1.9(a). □

So ℓ is a *potential* whose "Laplacian" is the vector of local log-abundancies, and the defect measures how far L is from being an in-degree Laplacian. For a k-perfect number, $\mathbf{1}^{\mathsf{T}} L = -v(k)^{\mathsf{T}}$.

Example 2.16 ($n = 6$)**.** In the order $(2, 3)$, $D = \operatorname{diag}(1, 1)$ and

$$L = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}, \qquad L\ell = \begin{pmatrix} \log \frac{2}{3} \\ \log \frac{3}{4} \end{pmatrix} = - \begin{pmatrix} \log h(2) \\ \log h(3) \end{pmatrix}.$$

The column sums are $(-1, 0) = -e_2$, and $\eta_2 + \eta_3 = \log(\frac{3}{2} \cdot \frac{4}{3}) = \log 2$.

Example 2.17 ($n = 120$)**.** In the order $(2, 3, 5)$,

$$L = \begin{pmatrix} 3 & -1 & -1 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}, \qquad L\ell = \begin{pmatrix} \log \frac{8}{15} \\ \log \frac{3}{4} \\ \log \frac{5}{6} \end{pmatrix} = - \begin{pmatrix} \log h(2^3) \\ \log h(3) \\ \log h(5) \end{pmatrix}.$$

The column sums are $(0, -1, 0) = -e_3$, and $\sum \eta_s = \log\bigl(\frac{15}{8} \cdot \frac{4}{3} \cdot \frac{6}{5}\bigr) = \log 3$.

Remark 2.18 (Taking logarithms loses nothing). The numbers $\log r$, r prime, are linearly independent over $\mathbb{Q}$: if $\sum_r c_r \log r = 0$ with rational c_r, clear denominators and exponentiate to get $\prod_r r^{mc_r} = 1$, so every $c_r = 0$. Passing from a multiplicative identity among prime powers to its logarithm is therefore injective as long as the coefficients stay rational. Information is lost only when the coefficients are relaxed to arbitrary reals, or when several primes are merged into one composite coordinate.

2.4 Harmonic morphisms and equitable partitions

Definition 2.19 (Harmonic morphisms)**.** A morphism $\varphi\colon G \to H$ of undirected multigraphs is *harmonic* if for every vertex x of G the number

$$m_\varphi(x) = \#\{e \ni x : \ \varphi(e) = e'\}$$

is the same for every edge e' of H at $\varphi(x)$. Edges that map to vertices are *vertical*.

See Urakawa [29], Baker–Norine [2], Joyner–Melles [16], and, for the weighted theory, the chapter "Weighted harmonic morphisms" of [18]. With $\Delta f(x) = \sum_{e=xy}(f(x) - f(y))$, the basic identity is

$$\Delta_G(f \circ \varphi)(x) = m_\varphi(x)\,(\Delta_H f)(\varphi(x)), \tag{B2}$$

because vertical edges contribute 0 and each edge e' at $\varphi(x)$ is covered exactly $m_\varphi(x)$ times. Baker and Norine also show that pullback and pushforward of divisors preserve principal divisors.

For a weighted digraph, a vertex quotient is given by a partition $V = C_1 \sqcup \cdots \sqcup C_k$. The harmonic condition becomes an equitability condition, and for a digraph there are two of them.

Definition 2.20 (Equitable partitions of a digraph)**.** A partition of V is

- *in-equitable* if for all cells C, C' the in-weight $\sum_{s\in C} w(s,r)$ is the same number $b_{CC'}$ for every $r \in C'$;
- *out-equitable* if for all cells C, C' the out-weight $\sum_{r\in C'} w(s,r)$ is the same number $b'_{CC'}$ for every $s \in C$.

The matrices $(b_{CC'})$ and $(b'_{CC'})$ are the corresponding quotient matrices.

For a symmetric W the two notions coincide with the weighted equitable partitions of the chapter "Edge-weighted graph partitions" of [18]: each vertex of a cell sends the same total weight to each cell. In matrix terms, an out-equitable partition is one whose cell indicator vectors span a W-invariant subspace; an in-equitable partition does the same for W^{T}.

Example 2.21 (Equitability is rare)**.** In the σ-graph of 120 (Example 2.12), try the partition $\{2\}, \{3,5\}$. The in-weight from $\{3,5\}$ into 3 is $w(3,3)+w(5,3) = 1$, but into 5 it is $w(3,5)+w(5,5) = 0$. So the partition is not in-equitable. The out-weight from 3 into $\{2\}$ is 2, but from 5 it is 1, so it is not out-equitable either.

2.5 Arithmetical structures

Definition 2.22 (Lorenzini)**.** Let $\Gamma = (V, E)$ be a connected (multi)graph with adjacency matrix A. An *arithmetical structure* on Γ is a pair of maps $R, S\colon V \to \mathbb{Z}_{>0}$ such that $\gcd\{R(v) : v \in V\} = 1$ and

$$R(v)\,S(v) = \sum_{w\sim v} R(w) \qquad (v \in V),$$

the sum counting multiple edges with multiplicity. Equivalently, $(\operatorname{diag}(S) - A)R = 0$. The matrix $\operatorname{diag}(S) - A$ is the *generalized Laplacian* of the structure, and the torsion subgroup of its cokernel $\mathbb{Z}^V/(\operatorname{diag}(S) - A)\mathbb{Z}^V$ is the *arithmetical critical group* $K(\Gamma; R, S)$.

Arithmetical structures are due to Lorenzini [20]. Every connected graph has the *natural* structure $R \equiv 1$, $S = \deg$, whose generalized Laplacian is the ordinary Laplacian and whose critical group is the usual critical (sandpile) group. Other structures exist as well; [18] gives the example $R = (1, 1, 2)$, $S = (3, 3, 1)$ on the triangle C_3. That source also shows that a harmonic morphism $\phi\colon \Gamma_2 \to \Gamma_1$ pulls an arithmetical structure (R_1, S_1) back to $(R_1 \circ \phi,\ \mu S_1 \circ \phi + \nu)$, where μ and ν are the horizontal and vertical multiplicities, and that the generalized Laplacians then satisfy $Q_2\Phi = D_\mu \Phi Q_1$.

Nothing in Definition 2.22 uses the symmetry of A. So it extends verbatim to weighted digraphs, in two ways.

Definition 2.23 (Arithmetical structures on a weighted digraph)**.** Let W be the weight matrix of a strongly connected digraph on V. A pair (R, S) of positive integer vectors with $\gcd R = 1$ is

- an *out-structure* if $S(s)R(s) = \sum_r w(s, r)R(r)$ for all s, i.e. $(\operatorname{diag}(S) - W)R = 0$;
- an *in-structure* if $S(r)R(r) = \sum_s w(s, r)R(s)$ for all r, i.e. $R^{\mathsf{T}}(\operatorname{diag}(S) - W) = 0$.

In both cases $\operatorname{diag}(S) - W$ is the generalized Laplacian, and the arithmetical critical group is the torsion of its cokernel. For a symmetric W both notions agree with Definition 2.22.

An in-structure on G is an out-structure on the reversed digraph. The pair $(\mathbf{1}, \text{in-degrees})$ is always an in-structure, the directed analogue of the natural structure. When S is the vector of in-degrees, the out-degree Laplacian of the reversed digraph is $Q' = \operatorname{diag}(S) - W^{\mathsf{T}}$, and the vector R of an out-structure is exactly the positive vector with $Q'^{\mathsf{T}}R = 0$. For strongly connected digraphs this vector is central to Asadi–Backman's chip-firing and Riemann–Roch theory for directed graphs [1].

Example 2.24 (The 2-cycle of 6)**.** For the digraph of Example 2.4, take S to be the in-degrees $(2, 1)$. The out-structure equations are $2R(2) = R(3)$ and $R(3) = 2R(2)$, so $R = (1, 2)$. The in-structure with the same S is $R = (1, 1)$. The two structures share the generalized Laplacian $\begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}$, with right kernel spanned by $(1, 2)$ and left kernel by $(1, 1)$.

2.6 Undirected encodings of a digraph

A digraph can be turned into an undirected graph in several ways. Some keep all the information; some do not.

Definition 2.25. Let G be a weighted digraph on V with weight matrix W.

- The *underlying graph* $U(G)$ is the simple graph with an edge $\{s, r\}$ whenever $w(s, r) > 0$ or $w(r, s) > 0$.
- The *source–target bipartite graph* $B(G)$ has two copies $V^- = \{s^-\}$ and $V^+ = \{r^+\}$ of V, and an undirected edge $s^- - r^+$ of weight $w(s, r)$. Its biadjacency matrix is W.
- The *symmetric dilation* of W is $\begin{pmatrix} 0 & W \\ W^{\mathsf{T}} & 0 \end{pmatrix}$, the weighted adjacency matrix of $B(G)$.

G is *symmetric* if $W = W^{\mathsf{T}}$. Then G is just an undirected multigraph with $w(s, r)$ parallel edges between s and r.

$B(G)$ loses nothing, provided its two color classes and labels are kept: the weighted degree of r^+ is the in-degree of r, and that of s^- is the out-degree of s. This correspondence between digraphs and balanced bipartite graphs is classical; see Brualdi–Harary–Miller [5]. The nonzero eigenvalues of the symmetric dilation are $\pm$ the singular values of W. By contrast $U(G)$ forgets both direction and weight.

Two facts about undirected graphs will be used for symmetric σ-graphs.

- **Handshake lemma.** The degrees of an undirected multigraph sum to twice the number of edges. So the number of odd-degree vertices is even.
- **Maximum principle.** For a Laplacian $\Delta f(x) = \sum_y w(x, y)(f(x) - f(y))$ with $w \geq 0$, at a vertex where f is largest $\Delta f \geq 0$, with strict inequality if some neighbor has a smaller value.

Example 2.26. For the σ-graph of 6, $U(G)$ is a single edge $\{2, 3\}$. The graph $B(G)$ has edges $2^- - 3^+$ of weight 1 and $3^- - 2^+$ of weight 2. The degree of 2^+ is 2, the in-degree of 2. The σ-graph of 6 is not symmetric, since $w(2, 3) = 1 \neq 2 = w(3, 2)$.

2.7 Divisor graphs and Hasse graphs

The σ-graph has *primes* as vertices. A more classical graph uses *divisors* as vertices. Divisor graphs have a literature of their own, surveyed by Ravi and Desikan [26].

Definition 2.27. Let $n = \prod_{i=1}^{k} p_i^{a_i} > 1$.

- The *divisor lattice* $\mathrm{Div}(n)$ is the set of positive divisors of n, partially ordered by divisibility. It is a lattice in the order-theoretic sense, with meet gcd and join lcm. Its bottom element is 1 and its top element is n.
- The *divisor graph* $D(n)$ is the comparability graph of $\mathrm{Div}(n)$: its vertices are the divisors, with distinct d, d' adjacent when $d \mid d'$ or $d' \mid d$.
- The *Hasse graph* $H(n)$ is the undirected graph of covering relations of $\mathrm{Div}(n)$: d and d' are adjacent when $d' = dp$ for a prime p. Coloring such an edge by p gives an edge-colored graph.

Terminology. These notes keep three objects apart.

- "Divisor lattice" always means the poset $\mathrm{Div}(n)$.
- "Hasse graph" means the undirected graph $H(n)$.
- "Labeled Hasse graph" means $H(n)$ together with the vertex weights $m(d) = d$.

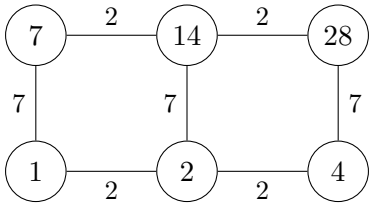
As an unlabeled graph, $H(n)$ does not know which corner is 1 and which is n, since reversing a path is a graph automorphism. The labels, or equivalently the order of $\mathrm{Div}(n)$, single out the bottom vertex 1 and the top vertex n. Words such as "top" and "upward" always refer to the divisibility order.

Writing a divisor as an exponent vector $(e_1, \ldots, e_k)$ with $0 \leq e_i \leq a_i$ identifies $H(n)$ with a Cartesian product of paths,

$$H(n) \cong P_{a_1+1} \,\square\, P_{a_2+1} \,\square \cdots \square\, P_{a_k+1},$$

in which the edges of color p_i are those in the ith coordinate direction.

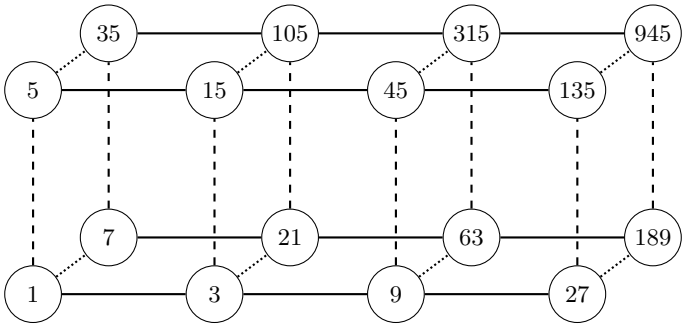
Example 2.28 ($D(6)$, $H(6)$ and $H(28)$)**.** The divisors of 6 are $1, 2, 3, 6$. Every pair is comparable except $\{2, 3\}$, so $D(6)$ is K_4 with one edge removed, and $H(6)$ is the 4-cycle $1{-}2{-}6{-}3{-}1 \cong P_2 \,\square\, P_2$. For $28 = 2^2 \cdot 7$, the graph $H(28) \cong P_3 \,\square\, P_2$ is a 2×3 grid. Horizontal edges have color 2 and vertical edges color 7:



The shape depends only on the exponents. The divisibility order of n is a product of chains whose lengths are the exponents. So, as unlabeled graphs, $D(n)$ and $H(n)$ depend only on the multiset of exponents of n, not on its primes.

Example 2.29 (One graph, three kinds of numbers)**.** The numbers $12 = 2^2 \cdot 3$, $28 = 2^2 \cdot 7$ and $45 = 3^2 \cdot 5$ all have exponent pattern $\{2, 1\}$. So $H(12) \cong H(28) \cong H(45)$ (the grid of Example 2.28), and likewise for D. Yet 12 is abundant, 28 is perfect, and 45 is deficient. Similarly $D(6) \cong D(15) \cong D(35)$, with only 6 perfect.

Example 2.30 (The Hasse graph $H(945)$)**.** For $945 = 3^3 \cdot 5 \cdot 7$ the graph $H(945) \cong P_4 \,\square\, P_2 \,\square\, P_2$ has $\sigma_0(945) = 16$ vertices. In the figure, solid edges multiply by 3, dashed edges by 5, and dotted edges by 7. The back layer is the front layer times 7.



The vertex labels add up to $\sigma(945) = 40 \cdot 6 \cdot 8 = 1920$, and the three coordinate chains have sums 40, 6 and 8. These are the local divisor sums whose factorizations give the σ-graph of Example 2.11.

Remark 2.31 (A terminology trap)*.* In graph theory a *perfect graph* is one in which every induced subgraph has chromatic number equal to its clique number. This has nothing to do with perfect numbers. In fact every divisor graph $D(n)$ is a perfect graph, because comparability graphs of partial orders are perfect.

The labeled Hasse graph and its partition function. Give the vertex $(e_1, \ldots, e_k)$ of $H(n)$ the weight $m(e) = \prod_i p_i^{e_i}$, the divisor it represents, and put $Z(n) = \sum_v m(v)$. Because $H(n)$ is a product of paths, Z factors:

$$Z(n) = \prod_{i=1}^{k} (1 + p_i + \cdots + p_i^{a_i}) = \prod_{i=1}^{k} \sigma(p_i^{a_i}) = \sigma(n).$$

For the grid of Example 2.28, $Z(28) = (1 + 2 + 4)(1 + 7) = 56$.

Harmonic projections. The coordinate projection $\pi_i \colon H(n) \to P_{a_i+1}$, $(e_1, \ldots, e_k) \mapsto e_i$, is a harmonic morphism. Each edge of the path at e_i has exactly one edge above it at every vertex of the fiber, so the horizontal multiplicity is 1; the edges in the other directions are vertical. Pushing m forward along π_i gives

$$(\pi_i)_* m\,(e) = p_i^e \prod_{j \neq i} \sigma\big(p_j^{a_j}\big).$$

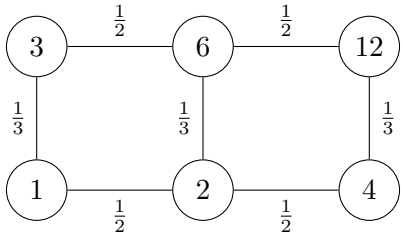
So the partition function of the labeled Hasse graph is assembled from the partition functions $\sigma(p_i^{a_i})$ of its coordinate paths. In $\mathrm{Div}(n)$ these paths are the chains $1, p_i, \ldots, p_i^{a_i}$, so $\sigma(p_i^{a_i})$ is called the ith *chain sum*.

An edge-weighted version. The vertex labels of $H(n)$ can be traded for edge weights. Give the edge from d to dp the weight $1/p$. Along any upward path from d to n, the product of the edge weights is d/n, whatever path is chosen. (Around each square d, dp, dq, dpq the two routes give the same product, and these squares generate the cycle space.) So the vertex labels are recovered from the edge weights, and

$$h(n) = \sum_{d \mid n} \frac{d}{n} = \sum_{v} \prod_{e \in \mathrm{path}(v \to n)} w(e).$$

In additive form, with edge length $\log p$ the distance from d to n is $\log(n/d)$, and $h(n) = \sum_v e^{-\operatorname{dist}(v,n)}$. This is an exact duality between vertex weights and edge weights. It adds no arithmetic information: the edge weights are a gradient of the vertex potential $\log d$.

Example 2.32 (Edge weights on $H(12)$)**.** For $12 = 2^2 \cdot 3$, horizontal edges multiply by 2 and get weight $\frac{1}{2}$; vertical edges multiply by 3 and get weight $\frac{1}{3}$:



From 2 up to 12 the weights multiply to $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} = \frac{2}{12}$, by either route. Summing over all six vertices gives $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{12} = \frac{7}{3} = h(12)$.

How the Hasse graph and the σ-graph fit together. The chain sums $\sigma(p_i^{a_i})$ are exactly the local divisor sums from which the σ-graph is built. The out-edges of p_i in $\widehat{G}_\sigma(n)$ record the prime factorization of the ith chain sum. So the two graphs play complementary roles.

- The labeled Hasse graph $H(n)$ records the *product structure*: $\sigma(n)$ is a product of chain sums.
- The σ-graph records the *interaction*: which primes occur in which chain sums, and how often.

Example 2.33 (28 in both languages)**.** In $H(28)$ the chain in the 2-direction has sum $1+2+4 = 7$, and the chain in the 7-direction has sum $1 + 7 = 8 = 2^3$. In $\widehat{G}_\sigma(28)$ these become the edges $2 \xrightarrow{1} 7$ and $7 \xrightarrow{3} 2$. Multiplying, $7 \cdot 2^3 = 2 \cdot 28$: the prime 7 returns once, matching its exponent 1, and the prime 2 returns three times, one more than its exponent 2. That extra factor of 2 is the defect $d(28) = e_2$.

3 Odd perfect numbers and their σ-graphs

From now on N denotes a hypothetical odd perfect number, so $\sigma(N) = 2N$. This section specializes the general constructions to N.

3.1 Euler's form and the vertex data

Euler showed that an OPN has the form

$$N = p^{\alpha} \prod_{q \in Q} q^{2\beta_q}, \qquad p \equiv \alpha \equiv 1 \pmod 4,$$

with $p \notin Q$ and all primes odd. The prime p is the *Euler* (or special) prime, and every $q \in Q$ is *nonspecial*. Correspondingly p^{α} is the *special component* and the $q^{2\beta_q}$ are the *nonspecial components*.

Definition 3.1 (Vertex data)**.** Put $S = \{p\} \cup Q$, the set of prime divisors of N. For $s \in S$ let

$$a_s = \begin{cases} \alpha, & s = p, \\ 2\beta_s, & s \in Q, \end{cases} \qquad n_s = a_s + 1.$$

Thus n_q is odd for $q \in Q$, while $n_p = \alpha + 1 \equiv 2 \pmod 4$. For nonspecial q we also write $e_q = n_q = 2\beta_q + 1$.

Note that $\omega(N) = |S|$ and $\Omega(N) = \sum_{s \in S} a_s$.

Local divisor sums. For each $s \in S$, $\sigma(s^{a_s}) = (s^{n_s} - 1)/(s - 1)$, and $\sigma(N) = \prod_{s \in S} \sigma(s^{a_s})$. The congruences in Euler's form give $v_2(\sigma(p^{\alpha})) = 1$, while each $\sigma(q^{2\beta_q})$ is odd. So we define the *odd local divisor sum*

$$\Sigma_s = \begin{cases} \sigma(p^{\alpha})/2, & s = p, \\ \sigma(q^{a_q}), & s = q \in Q, \end{cases} \qquad \text{so that} \qquad \prod_{s \in S} \Sigma_s = N.$$

Definition 3.2 (Exponent excess)**.** $E(N) = \Omega(N) - (2\omega(N) - 1) = (\alpha - 1) + \sum_{q \in Q}(2\beta_q - 2)$. It vanishes exactly when $\alpha = 1$ and every $\beta_q = 1$.

3.2 The σ-graph G_N

The extended σ-graph of an OPN has exactly one boundary vertex, the prime 2. It receives a single edge, of weight 1, from the Euler prime. Deleting it gives the graph used throughout the rest of these notes.

Definition 3.3 (The σ-graph of an OPN)**.** The *σ-graph* G_N is the weighted digraph on S with

$$w(s, r) = v_r\big(\Sigma_s\big) = v_r\big(\sigma(s^{a_s})\big) \qquad (s, r \in S).$$

Write $W = (w(s, r))_{s,r \in S}$ and $D = \operatorname{diag}(a_s)$.

So G_N is the internal σ-graph of N, and also the extended σ-graph with the vertex 2 removed. For odd r the factor 2 in Σ_p is irrelevant. Every weight is computed by Proposition 1.25.

Definition 3.4 (Logarithmic quantities)**.** Let $\ell \in \mathbb{R}^S$ be the log-prime vector, $\ell_s = \log s$. Define

$$\delta_s = \log \Sigma_s - a_s \log s,$$

the logarithm of the odd local abundancy. For $A, B \subseteq S$ define the *logarithmic flow*

$$L(A, B) = \sum_{s \in A} \sum_{r \in B} w(s, r) \log r.$$

For $q \in Q$, $\delta_q = \eta_q = \log h(q^{a_q})$. For the Euler prime, $\delta_p = \eta_p - \log 2$, which accounts for the deleted vertex 2.

Definition 3.5 (Cyclotomic obligations)**.** For $q \in Q$ and $d \mid e_q$ with $d > 1$, the pair (q, d) is a *cyclotomic obligation*. Its set of possible targets is

$$P(q, d) = \{r \in S : \ \mathrm{ord}_r(q) = d\}.$$

Write $\mathcal{O}$ for the set of all obligations.

By Proposition 1.20, the obligations at q are the cyclotomic layers of $\sigma(q^{a_q})$; there are $\sigma_0(e_q) - 1$ of them.

Definition 3.6 (Edge terminology)**.** Let $q \in Q$ and suppose $q \to r$. The edge is an *order-one edge* if $\mathrm{ord}_r(q) = 1$. If $d = \mathrm{ord}_r(q) > 1$, it is an *order-$d$ edge*; equivalently, r is a primitive prime divisor of $q^d - 1$. When the obligation in question is (q, d), such an edge is also called a *primitive edge of level d*. A *multiplier contribution* is not a separate edge type: it is the extra valuation $v_r(n_q/d)$ that can sit on an order-d edge when $r \mid n_q/d$.

3.3 Defect and Poisson forms of the OPN conjecture

Proposition 2.13 gives the first reformulation at once.

Corollary 3.7 (Defect form of the OPN conjecture)**.** *There is an odd perfect number if and only if there is an odd $n > 1$ whose extended σ-graph has defect vector e_2. That is, the boundary vertex 2 receives total weight exactly 1, and every other vertex receives total weight exactly equal to its exponent.*

For odd n the vertex 2 is always a boundary vertex, and $d_2(n) = v_2(\sigma(n))$ is the total power of 2 contributed by the local divisor sums. So Corollary 3.7 repackages two classical facts: exactly one local divisor sum is even (this is Euler's special prime), and it is $\equiv 2 \pmod 4$. The odd abundant number 945 fails badly: all three of its local divisor sums are even (Example 2.14).

Deleting the boundary vertex turns the general Poisson equation of Proposition 2.15 into a cleaner one.

Proposition 3.8 (The Poisson equation of an OPN)**.** *For an odd perfect number, with $L = D - W$ on S,*

$$\boxed{L\ell = -\delta, \qquad \mathbf{1}^{\mathsf{T}} L = 0}$$

where $\delta_q > 0$ for $q \neq p$ and $\delta_p < 0$.

Proof. Row s of $L\ell$ is $a_s \log s - \log \Sigma_s = -\delta_s$. Column r sums to $a_r - v_r(N) = 0$, because $\prod_s \Sigma_s = N$. The signs come from Lemma 1.9(b): $\Sigma_q = \sigma(q^{a_q}) > q^{a_q}$, while $\sigma(p^\alpha)/p^\alpha < p/(p-1) \leq 5/4 < 2$. □

So L has zero column sums: it is an in-degree Laplacian. The log-prime vector is a *potential* for the real degree-zero vector δ, which is positive everywhere except at the Euler prime. This is formally analogous to a principal divisor with potential log, but it is not literally the Baker–Norine notion. Baker–Norine theory uses undirected graphs, integral divisors and integral firing functions; here the operator is directed and ℓ is real-valued.

Two quick consequences:

- $\sum_s \delta_s = \mathbf{1}^\mathsf{T}(-L\ell) = 0$. This is the Fredholm compatibility condition for the left null vector $\mathbf{1}^\mathsf{T}$. For a general directed Laplacian there may be more left null vectors. For an OPN there are not: G_N is strongly connected (Proposition 4.11), so L has rank $|S| - 1$ (Corollary 4.13).
- Summing the equation over a set T gives the cut identity (L5) of Section 4.

Constant vectors need not satisfy $L\mathbf{1} = 0$: $L\mathbf{1} = (a_s - \Omega(\Sigma_s))_s$. Its entries sum to 0, since both $\sum_s a_s$ and $\sum_s \Omega(\Sigma_s)$ equal $\Omega(N)$, but nothing forces them all to vanish. This vector reappears as the "potential" in the quadratic-form identity of Section 6.6.

The OPN conjecture is therefore equivalent to:

> *There is no finite set S of odd primes with Euler-form exponents a_s such that the integer matrix W defined by $w(s,r) = v_r(\Sigma_s)$ satisfies $\mathbf{1}^\mathsf{T}(D - W) = 0$ and σ-closure.*

Given those, $L\ell = -\delta$ is automatic.

3.4 What is general and what is forced

The examples of Section 2 show which features of G_N are special to perfectness.

- **General.** The order/LTE classification of edge weights (Proposition 1.25), the cyclotomic layers (Proposition 1.20), primitive prime divisors (Theorem 1.23), and the identities $b_r = v_r(\sigma(n))$ and $L\ell = -\eta$ hold for every integer.
- **Forced by perfectness.** Exact balance $b = a$ away from the prime 2, σ-closure (no boundary vertex except 2), the orientation of the logarithmic flow toward the Euler prime (Lemma 4.9), and strong connectivity (Proposition 4.11) all fail in general. The numbers 15 and 945 show this.

Natural questions for arbitrary n include which integers have strongly connected internal σ-graphs, how the defect vector separates abundant from deficient numbers, and how $\widehat{G}_\sigma(n)$ changes when n is multiplied by a new prime power. The known even perfect and multiperfect numbers are the natural test cases.

4 The proposed program, with examples

This section restates the AI-generated program described in the abstract. That program builds on the 2026 preprint of Jeremy Rodgers [27], an early-stage, non-peer-reviewed paper that introduces the same weighted digraph (there called $G_\sigma(N)$, with edge weights $w_{s\to r} = v_r(\sigma(s^{a_s}))$). Several of the lemmas below have direct counterparts in that preprint:

- valuation balance (Lemma 4.2) is its Theorem 3.2 and Proposition 4.1;
- the order/LTE classification (Lemma 4.3) corrects its Theorem 5.1 and extends its Theorem 5.2 to the order-one case;

- the primitive-divisor count $\sigma_0(e_q) \le \omega(N)$ (Lemma 4.4) is its Proposition 8.2 and Theorem 9.3;
- the maximal-prime statement (Lemma 4.14) is its Theorem 7.2, and its §8.3 gives the related count $\#\{q : \mathrm{ord}_R(q) = d\} \le \varphi(d)$;
- the cheap-edge bound (Lemma 4.15) is its Proposition 4.2/8.1.

The logarithmic cut identity, the orientation lemma, and the two "breakthrough" claims do not appear in the preprint; they were added by the AI-generated post.

4.1 The correction to the order criterion

Theorem 5.1 of the preprint states, for distinct odd primes q, r and $e = 2\beta + 1$,

$$r \mid \sigma(q^{2\beta}) \iff d := \mathrm{ord}_r(q) > 1 \text{ and } d \mid e.$$

The implication "$\Leftarrow$" is correct. The implication "$\Rightarrow$" is false: it says that a prime dividing $q-1$ can never divide $\sigma(q^{2\beta})$. The proof in the preprint asserts that $r \mid (q^e-1)/(q-1)$ holds exactly when $r \mid q^e - 1$ and $r \nmid q-1$, which overlooks the possibility that r divides $q-1$ *and* divides q^e-1 to a higher power. The smallest example already shows the phenomenon:[2]

Example 4.1. $\sigma(13^2) = 1 + 13 + 169 = 183 = 3 \cdot 61$, but $13 \equiv 1 \pmod 3$, so $\mathrm{ord}_3(13) = 1$. The factor 3 survives the division by $13 - 1$ because

$$v_3(13^3 - 1) = v_3(2196) = v_3(2^2 \cdot 3^2 \cdot 61) = 2 > 1 = v_3(12).$$

This is not merely a technicality. An edge $q \to r$ with $\mathrm{ord}_r(q) = 1$ is exactly the situation in which r divides both $q-1$ and the exponent e_q; it is the mechanism behind the classical observation that 3 divides $\sigma(q^2)$ whenever $q \equiv 1 \pmod 3$. Lemma 4.3 below treats it as a separate case.

4.2 Lemmas 1–8

The lemmas below are stated for an OPN N with the notation of Section 3.

Lemma 4.2 (Exact valuation balance)**.** *For every $r \in S$,*

$$a_r = \sum_{s \in S} w(s, r). \tag{L1}$$

Moreover every odd prime dividing some Σ_s lies in S (σ-closure).

Proof. Compare v_r of both sides of $\prod_s \Sigma_s = N$. □

In matrix form, (L1) says $\mathbf{1}^{\mathsf{T}} W = (a_s)_s$: the weighted *in*-degree of r is a_r. There is no matching statement about out-degrees, so this is not ordinary flow conservation.

[2] Counterexamples of this kind are common, not exotic. *Every* prime $q \equiv 1 \pmod 3$ appearing to exponent 2 gives one, since then $q^2 + q + 1 \equiv 3 \pmod 9$; for example $\sigma(7^2) = 3 \cdot 19$ and $\sigma(19^2) = 3 \cdot 127$. Among all primes $q < 200$ and exponents $2\beta \le 8$, 58 of the 497 prime factors r of $\sigma(q^{2\beta})$ satisfy $\mathrm{ord}_r(q) = 1$, about 12%. The correct criterion, which follows from Lemma 4.3 below, is

$$r \mid \sigma(q^{2\beta}) \iff \big(\mathrm{ord}_r(q) > 1 \text{ and } \mathrm{ord}_r(q) \mid e\big) \text{ or } \big(\mathrm{ord}_r(q) = 1 \text{ and } r \mid e\big).$$

The same faulty criterion is restated in §10.1 and Appendix B.3 of the preprint.

How to read (L1). If $r^2 \parallel N$, then the total number of copies of r supplied by *all the other* local divisor sums must be exactly two. This could happen through two edges of weight 1, through one edge of weight 2, or through some other partition of the integer 2; what cannot happen is a total incoming weight 1 or 3. Thus an exponent in N is simultaneously a *demand* that the rest of the graph must meet. This very elementary observation is the source of most of the "capacity" language below.

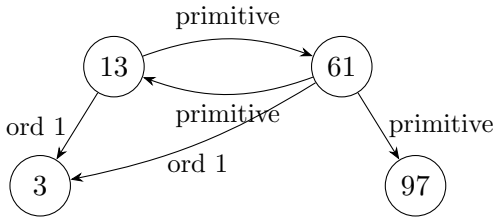
Lemma 4.3 (Order/LTE classification of edges)**.** *For distinct $s, r \in S$, the weight $w(s,r) = v_r(\sigma(s^{a_s}))$ is given by Proposition 1.25 with $a = a_s$ and $n = n_s$. In particular $w(s,r) = 0$ unless $\mathrm{ord}_r(s) \mid n_s$; $w(s,r) = v_r(n_s)$ when $\mathrm{ord}_r(s) = 1$; and $w(s,r) = v_r(s^d - 1) + v_r(n_s/d)$ when $d = \mathrm{ord}_r(s) > 1$ divides n_s.* *(L2a), (L2b)*

The three sources of weight, with examples, are discussed after Proposition 1.25. For the edge $13 \to 3$ of Example 4.1, Case II gives $w(13,3) = v_3(3) = 1$.

Lemma 4.4 (Cyclotomic obligations)**.** *For $q \in Q$, $\sigma(q^{a_q}) = \prod_{d \mid e_q,\, d>1} \Phi_d(q)$. Each obligation (q,d) has a primitive prime $\ell_{q,d} \in P(q,d) \subseteq S$ with $\ell_{q,d} \equiv 1 \pmod{2d}$. Distinct d give distinct $\ell_{q,d}$, so $\sigma_0(e_q) \le \omega(N)$.*

Zsigmondy does *not* say $\ell_{q,d}$ is a new prime; an existing vertex may be recycled. Note also that the bound $\sigma_0(e_q) \le \omega(N)$ controls the number of *layers*, not the size of the exponent: a prime value of e_q has only one layer, however large it is.[3]

Example 4.5 (A primitive 2-cycle)**.** 61 is primitive for $13^3 - 1$ (Example 4.1), and $61^2 + 61 + 1 = 3783 = 3 \cdot 13 \cdot 97$ with $\mathrm{ord}_{13}(61) = 3$ (indeed $61 \equiv 9$ and $9^3 = 729 \equiv 1 \pmod{13}$). So if $a_{13} = a_{61} = 2$, the σ-graph contains the primitive edges $13 \to 61$ and $61 \to 13$:



All edges have weight 1. ($61 \equiv 1 \pmod 3$, so $61 \to 3$ is an order-one edge.)

Example 4.6 (A layer that folds downward)**.** $\Phi_3(67) = 67^2 + 67 + 1 = 4557 = 3 \cdot 7^2 \cdot 31$: every prime factor is smaller than 67.

Lemma 4.7 (Obligation capacity)**.** *For each $r \in S$,*

$$\#\{q \in Q : \ \mathrm{ord}_r(q) > 1,\ \mathrm{ord}_r(q) \mid e_q\} \le a_r, \tag{L4}$$

and globally $\sum_{q \in Q} (\sigma_0(e_q) - 1) \le \Omega(N)$.

[3] This point is overlooked in Rodgers' Proposition 9.4, which asserts that a bound on the number of divisors $d \mid e(q)$ whose primitive prime lies in S would bound $e(q)$ itself. Every prime e is a counterexample, since then $\sigma(q^{e-1}) = \Phi_e(q)$ is a single layer. For instance $\sigma(3^{12}) = \Phi_{13}(3) = 797161$ is itself prime: one layer, one primitive prime, and the same is true in shape for every prime exponent e. The preprint's own §9.3 makes the correct observation that small $\sigma_0(e)$ is compatible with large e, so the proposition contradicts the discussion that precedes it.

Why this is a capacity statement. Every source q counted on the left contributes at least one copy of r to $\sigma(q^{a_q})$, hence at least one unit to the incoming weight at r. But Lemma 4.2 says that the *entire* incoming budget at r is only a_r. Therefore at most a_r different sources can each demand even one unit from r. The inequality deliberately throws away multiplicity and is consequently quite coarse.

For example,

$$\sigma(3^4) = 121 = 11^2,$$

so the single obligation $(3, 5)$ lands at 11 with weight 2. In the Hall count it consumes only one "obligation slot," even though arithmetically it uses two units of 11-adic capacity. This illustrates both the usefulness and the weakness of the unweighted obligation count: it detects overcrowding of targets, but it forgets how expensive some assignments actually are.

Lemma 4.8 (Logarithmic cut identity)**.** *$\delta_q > 0$ for $q \in Q$, $\delta_p < 0$, $\sum_s \delta_s = 0$, and for every $T \subseteq S$,*

$$\sum_{s \in T} \delta_s = L(T, S \setminus T) - L(S \setminus T, T). \tag{L5}$$

Proof. $\delta_q > 0$ since $\sigma(q^a) > q^a$. For p: $\sigma(p^\alpha)/p^\alpha < p/(p-1) \le 5/4 < 2$ since $p \ge 5$. $\sum \delta_s = \log \prod \Sigma_s - \log N = 0$. For (L5), write $\log \Sigma_s = \sum_r w(s, r) \log r$ and $a_r = \sum_s w(s, r)$ by (L1):

$$\sum_{s \in T} \delta_s = L(T, S) - \sum_{r \in T} a_r \log r = L(T, S) - L(S, T),$$

and the edges inside T cancel. □

Lemma 4.9 (Orientation toward the Euler prime)**.** *If $\varnothing \ne T \subseteq S \setminus \{p\}$, then $L(T, S \setminus T) > L(S \setminus T, T)$. Consequently every $q \in S$ has a directed path to p, and the condensation of G_N has a unique sink strong component, which contains p.*

Proof. The inequality is (L5) with all $\delta_q > 0$. If T is the set of vertices with no path to p, then T has no out-edges, so $L(T, S \setminus T) = 0 \le L(S \setminus T, T)$, forcing $T = \varnothing$. □

A singleton example. Take a nonspecial component 13^2. Its local divisor sum is $\sigma(13^2) = 183 = 3 \cdot 61$, so its outgoing logarithmic mass is $\log 3 + \log 61 = \log 183$, whereas its exponent demand is $2 \log 13 = \log 169$. Their difference is

$$\delta_{13} = \log \frac{183}{169} > 0.$$

Thus, if 13^2 occurred in an OPN, the singleton cut $T = \{13\}$ would export slightly more logarithmic mass than it imports. Lemma 4.9 says that the same phenomenon holds for *every* set consisting only of nonspecial vertices: after all internal exchanges cancel, that set must have a net logarithmic export. The Euler prime is therefore not merely a convenient distinguished vertex; it is the only place where the accumulated surplus can finally be absorbed.

Remark 4.10 (A directed Matrix–Tree consequence)*.* Lemma 4.9 says that every vertex can reach the Euler prime p. Consequently the support digraph of G_N contains a directed spanning in-arborescence rooted at p: for example, choose from each $q \ne p$ one edge that decreases the directed distance to p. The directed Matrix–Tree theorem [11] therefore gives a positive weighted cofactor for the *out-degree* Laplacian $L_{\text{out}} = D_{\text{out}} - W$ corresponding to such rooted arborescences. This is a canonical graph invariant suggested by the distinguished Euler sink. It should not, however, be confused with the arithmetic operator $D - W$ used later: there $D = \operatorname{diag}(a_s)$ is the in-degree/demand matrix supplied by perfectness, not the out-degree matrix. Any use of Matrix–Tree cofactors in the OPN problem therefore needs an arithmetic bridge between these two diagonals.

Reaching p versus being reached from p. Lemma 4.9 says that every vertex can *reach* the Euler prime. It is natural to ask about the reverse direction: can every vertex be reached *from* p? If so, G_N is strongly connected. The logarithmic identities alone do not settle this, because a set A that receives no edges from outside could still export a positive amount of logarithmic mass, which is perfectly consistent with (L5). The missing ingredient is integrality. The following result is not part of the original post; it is added here because it answers a question raised by the graph formulation itself.

Proposition 4.11 (Strong connectivity)**.** *For every nonempty proper subset $A \subsetneq S$ there is an edge of G_N from $S \setminus A$ into A. Equivalently, G_N is strongly connected.*

Proof. Suppose no edge enters A. Then every $t \in A$ receives its whole demand a_t from inside A, i.e. $a_t = \sum_{s \in A} w(s,t)$. Let $N_A = \prod_{t \in A} t^{a_t}$ be the corresponding unitary divisor. Then

$$\prod_{s \in A} \Sigma_s = \prod_{t \in S} t^{\sum_{s \in A} w(s,t)} = N_A \cdot M, \qquad M = \prod_{t \notin A} t^{\sum_{s \in A} w(s,t)} \in \mathbb{Z}.$$

Since $\sigma(N_A) = \prod_{s \in A} \sigma(s^{a_s})$ equals $2 \prod_{s \in A} \Sigma_s$ if $p \in A$ and $\prod_{s \in A} \Sigma_s$ otherwise, the abundancy $h(N_A) = \sigma(N_A)/N_A$ is an *integer*, and it exceeds 1 because $A \neq \varnothing$. On the other hand h is multiplicative on coprime arguments, so

$$2 = h(N) = h(N_A)\, h(N/N_A),$$

and $h(N/N_A) > 1$ because $A \neq S$. Hence $1 < h(N_A) < 2$, which is impossible for an integer. Finally, a digraph is strongly connected exactly when every nonempty proper vertex set has an entering edge (take A to be the set of vertices not reachable from a fixed vertex). □

Remark 4.12 (Why the argument needs $k = 2$). The proof uses only that no integer lies strictly between 1 and 2. For the even perfect numbers the conclusion is visible directly: the graph on $\{2, M_p\}$ is the directed 2-cycle $2 \rightleftarrows M_p$ (see Example 2.8). For a k-perfect number with $k \geq 3$ the same argument only shows that a closed set A would give a unitary divisor with integral abundancy $h(N_A) < k$, which is not a contradiction by itself.

The Euler prime is the unique *sink* of the logarithmic flow (Lemma 4.9), but Proposition 4.11 shows that it is not a sink of the *graph*: there are edges out of p leading back to every vertex. Strong connectivity has a pleasant linear-algebra consequence.

Corollary 4.13 (Rank of the Poisson operator)**.** *Let $L = D - W$ with $D = \operatorname{diag}(a_s)$. Then the left nullspace of L is spanned by $\mathbf{1}$, and $\operatorname{rank} L = \omega(N) - 1$. Consequently $Lx = b$ is solvable exactly when $\mathbf{1}^{\mathsf{T}} b = 0$, and its solution is unique up to adding a vector in the one-dimensional kernel of L.*

Proof. A row vector y^{T} satisfies $y^{\mathsf{T}} L = 0$ iff

$$y_r = \sum_{s \in S} \frac{w(s,r)}{a_r}\, y_s \qquad (r \in S),$$

i.e. $y = My$ with $M = D^{-1} W^{\mathsf{T}}$. By (L1) every row of M sums to 1, so M is the transition matrix of a Markov chain that walks *backward* along the edges of G_N, and y is a harmonic function for it. By Proposition 4.11 this chain is irreducible. At a vertex where y attains its maximum, y_r is an average of the values at its backward neighbors, so all of them are maximal too; by irreducibility the maximum spreads to every vertex and y is constant. Hence the left nullspace is $\mathbb{R}\mathbf{1}$ and the rank is $|S| - 1$. □

For example, $\ell = (\log s)_{s\in S}$ is pinned down by the equation $L\ell = -\delta$ of Section 3.3 up to a single additive direction. This settles a point that would otherwise need a separate hypothesis: the Fredholm condition $\sum_s \delta_s = 0$ is not merely necessary but sufficient for solvability. The additive direction itself, the kernel of L, turns out to carry interesting integer structure: it is an *arithmetical structure* in the sense of Lorenzini. Section 6.5 develops this.

Lemma 4.14 (Maximal prime)**.** *Let $R = \max S$. If $q \in Q$, $q < R$, supplies R, then $R \nmid e_q/\operatorname{ord}_R(q)$, so $w(q,R) = v_R(q^{\operatorname{ord}_R(q)} - 1)$. Also the number of such suppliers is at most $h_R - 1$, where $h_R = (R-1)/2^{v_2(R-1)}$ is the order of the odd-order subgroup of $(\mathbb{Z}/R\mathbb{Z})^\times$.* *(L7), (7.1)*

Why the largest prime is special. For a general target r, an edge can gain extra weight from the multiplier term $v_r(e_q/\operatorname{ord}_r(q))$ in Lemma 4.3. At the largest prime R this mechanism is severely constrained. If R divided that multiplier, then $e_q \geq R\operatorname{ord}_R(q) > R$. Zsigmondy applied at the larger exponent e_q would force a primitive divisor $\ell \equiv 1 \pmod{2e_q}$, hence $\ell > R$, contradicting maximality. Thus incoming weight at R has to come from the base-order term rather than from arbitrarily repeated multiplier lifting.

A small numerical model shows the idea. Since

$$\sigma(5^2) = 31,$$

we have $\operatorname{ord}_{31}(5) = 3$, so an exponent with $e_5 = 3$ can supply the target $R = 31$. But if one tried to obtain an extra factor of 31 from a multiplier by taking, say, $e_5 = 3 \cdot 31 = 93$, then a primitive divisor of $5^{93} - 1$ would be at least $2 \cdot 93 + 1 = 187 > 31$. Such a prime could not fit in a prime set whose maximum is 31.

The subgroup bound has a similar interpretation. Any supplier of R has odd multiplicative order modulo R, so its residue lies in the odd-order subgroup of $(\mathbb{Z}/R\mathbb{Z})^\times$. For the toy value $R = 31$, this subgroup has order $h_R = 15$; excluding the identity residue leaves at most 14 possible nontrivial supplier residues. For actual OPN bounds R is enormous, so this count is usually far too weak numerically, but it explains the group-theoretic content of the lemma.

Lemma 4.15 (Cheap-edge capacity)**.** *If all weights were 0 or 1, then $\Omega(N) = \sum_{s,r} w(s,r) \leq \omega(N)(\omega(N) - 1)$.*

Concrete meaning. A loopless digraph on k vertices has at most $k(k-1)$ directed edges. If every edge carries only one unit of valuation, the total incoming valuation available in the whole graph is therefore at most $k(k-1)$. But by Lemma 4.2 that total is exactly $\Omega(N)$.

For instance, Nielsen's theorem permits the extremal possibility $\omega(N) = 10$, but then a 0/1-weighted σ-graph could carry at most $10 \cdot 9 = 90$ units. The published bound $\Omega(N) \geq 101$ already exceeds 90. Consequently, if an OPN with exactly ten distinct prime factors existed, *some edge would necessarily have weight at least* 2. In arithmetic language, at least one local divisor sum would have to contain the square of another prime divisor of N. The limitation is that $k(k-1)$ grows quadratically while known lower bounds for Ω grow only linearly in k, so this neat contradiction does not persist for arbitrary k: already for $k = 11$ the capacity is $110 \geq 101$.[4]

[4]Rodgers' preprint gets this comparison wrong in two opposite ways. Its §8.1 says that $\Omega(N) \geq 101$ is compatible with the cheap-only bound "for the known ranges of $\omega(N)$," which fails at $\omega(N) = 10$, where the capacity is $90 < 101$. Its Appendix C.4 says that $\omega(N) \geq 10$ makes the capacity equal to 90 and so "rules out the pure cheap-only scenario," which fails for every $\omega(N) \geq 11$. The correct statement is the one in the text: an expensive edge is forced when $\omega(N) = 10$, and nothing is forced by this count alone when $\omega(N) \geq 11$.

The post compares this with published bounds: $\omega(N) \geq 10$ (Nielsen [21]), $\Omega(N) \geq 101$ (Ochem–Rao [22]), and $\Omega(N) \geq 2\omega(N) + 51$ (Ochem–Rao [23]). Ochem's current computational project page reports that the searches have since been pushed to $N > 10^{2200}$ and $\Omega(N) \geq 115$ [24]; these should be distinguished from the older published 10^{1500} and 101 theorems. A linear lower bound on Ω can never beat the quadratic cheap-edge capacity by itself, so counting edges alone will not suffice.

4.3 The two "breakthrough lemmas"

The following two statements were proposed as possible endgames. They are called "breakthrough" lemmas because either one, if proved under the stated hypotheses, would rule out OPNs. That is also the reason for caution: the hypotheses already encode so much of perfectness that the statements are closer to reformulations of the original problem than to presently accessible lemmas. It is nevertheless useful to spell them out, because they identify *what kind of obstruction* a graph-theoretic proof would need.

Claimed lemma 4.16 (A: Cyclotomic Hall deficiency)**.** For every Euler-form prime set satisfying σ-closure, (L1) and (L2), there is a family $\mathcal{A} \subseteq \mathcal{O}$ with

$$|\mathcal{A}| > \sum_{r \in U(\mathcal{A})} a_r, \qquad U(\mathcal{A}) = \bigcup_{(q,d) \in \mathcal{A}} P(q,d). \tag{H}$$

What Claim A is trying to say. Think of the cyclotomic obligations (q, d) as jobs that must be assigned to prime vertices. Zsigmondy guarantees that every job has at least one primitive-prime target. A target r has capacity a_r, because each job assigned to r contributes at least one unit to the incoming valuation at r, and the total incoming valuation is exactly a_r by (L1). Thus the primitive-prime incidence relation gives a bipartite matching problem with capacitated right-hand vertices.

Hall's theorem says that such an assignment can exist only if every family of jobs has enough total neighboring capacity. In the present notation the necessary inequality is

$$|\mathcal{A}| \leq \sum_{r \in U(\mathcal{A})} a_r.$$

Claim A asserts that the arithmetic always forces some family for which the reverse strict inequality holds. Such a family would be a certificate that the cyclotomic obligations cannot all be absorbed by the available primes.

Example 4.17 (A toy Hall obstruction)**.** Suppose five obligations $A_1, \ldots, A_5$ can land only at two nonspecial prime vertices r_1, r_2, each carrying the minimal exponent $a_{r_1} = a_{r_2} = 2$. Then their neighborhood has total capacity 4, while there are 5 obligations:

$$5 > 2 + 2.$$

No assignment is possible, regardless of how the five jobs are routed. This is exactly the combinatorial shape Claim A seeks inside an OPN. The example is schematic—actual target sets are determined by multiplicative orders—but its capacities are compatible with Euler form.

The arithmetic version is subtler because target sets usually overlap in complicated ways and capacities need not be 1. For example,

$$\Phi_5(3) = 121 = 11^2.$$

The obligation $(3, 5)$ has primitive target 11. Hall's count records one job assigned to a vertex of capacity a_{11}, while the actual factorization uses two units of 11-adic valuation. Thus the unweighted Hall model is a *relaxation*: it intentionally forgets some expensive multiplicity.

Why the stated hypothesis is too ambitious. An actual OPN already satisfies σ-closure and exact valuation balance. Therefore proving that *every* such configuration violates Hall is, in effect, proving that no OPN exists. A more realistic intermediate target is to impose only a selected family of obligations and derive a structural inequality. For example, one could use only the level-3 obligations from vertices with $a_q = 2$, where

$$\Phi_3(q) = q^2 + q + 1,$$

and try to prove that too many such low-exponent vertices cannot share a small set of targets. A theorem of the schematic form

$$\#\{q : a_q = 2\} \leq C\,\#\{r : a_r \geq 4\} + C_0$$

would not prove nonexistence, but it would immediately yield a new lower bound on Ω/ω. This is the kind of weakened Hall statement that could feed the discharging program in Section 6.

Claimed lemma 4.18 (B: Recycling capacity)**.** Every σ-closed, valuation-balanced Euler-form configuration satisfies $E(N) \leq 51$.

What exponent excess measures. The smallest exponents compatible with Euler's form are

$$\alpha = 1, \qquad a_q = 2 \quad (q \in Q).$$

If $k = \omega(N)$, this baseline has

$$\Omega_{\min} = 1 + 2(k - 1) = 2k - 1,$$

so $E(N) = \Omega(N) - (2\omega(N) - 1)$ measures exactly how far the exponent pattern lies above the minimal Euler pattern. Raising one nonspecial exponent from 2 to 4 adds 2 to E; raising the Euler exponent from 1 to 5 adds 4.

For example, with ten distinct prime factors the minimal Euler pattern would have $\Omega = 19$ and $E = 0$. If one nonspecial component had exponent 4 instead of 2, then $\Omega = 21$ and $E = 2$. At the opposite extreme, a single nonspecial exponent 54 in place of 2 would contribute the full 52 units of excess by itself. More generally, across a larger prime set, twenty-six separate increases from exponent 2 to exponent 4 would also contribute 52.

Ochem–Rao's published inequality

$$\Omega(N) \geq 2\omega(N) + 51$$

is exactly the statement

$$E(N) \geq 52.$$

Thus Claim B, which says $E(N) \leq 51$, would contradict a known theorem by one unit and finish the OPN problem.

Where "recycling" enters. Increasing an exponent changes $e_q = a_q + 1$ and usually creates additional cyclotomic layers $\Phi_d(q)$. Zsigmondy often forces primitive prime factors for those layers. The hoped-for charging argument is that a finite prime set cannot keep reusing the same vertices to absorb all of these new requirements without exhausting their valuation capacities. The difficulty is that divisor counts grow slowly, primitive primes can be recycled across different bases, and high-weight edges can absorb more than one valuation unit at a time. No known argument turns this intuition into the universal constant 51.

A more realistic weaker target. Instead of bounding $E(N)$ by an absolute constant, try to control how much excess can be supported by a specified number of high-exponent vertices. For example, let

$$H = \#\{q \in Q : a_q \geq 4\}.$$

An inequality of the form

$$\#\{q \in Q : a_q = 2\} \leq CH + C_0$$

would already force a positive proportion of nonspecial vertices to have exponent at least 4, and hence improve a lower bound for Ω/ω. Restricted-exponent theorems provide evidence that such questions can be tractable: Ochem–Zelinsky's restrained case, discussed in Section 5, assumes that all but one nonspecial exponent are 2 and obtains an extraordinarily strong restriction on the exceptional component.

Logical status. Claim A and Claim B should therefore be read as *targets that describe a possible final obstruction*, not as established reductions of the OPN conjecture. The educational value of the graph model is that it suggests weaker quantitative versions—capacity inequalities, forbidden local configurations, and bounds on the proportion of low-exponent vertices—that can be attacked without solving the whole problem at once.

5 Assessment

5.1 What is correct

Every numerical example and every proof sketch in Section 4 checks out. Example 4.1 is a genuine counterexample to Theorem 5.1 of Rodgers [27] as stated there. The other results that the program takes from the preprint—valuation balance, the LTE decomposition for edges of order greater than 1, primitive edges, the layer count $\sigma_0(e_q) \leq \omega(N)$, and the maximal-prime statement—are correct. Lemma 4.3, which adds the missing order-one case, is the right foundation for any work with σ-graphs.

5.2 Where the novelty is overstated

- **Lemma 4.8 is multiplicativity of abundancy.** For $T \subseteq S$ let $N_T = \prod_{s \in T} s^{a_s}$ be the corresponding unitary divisor. Then $\sum_{s \in T} \delta_s = \log h(N_T)$, with the factor 2 removed from p's term if $p \in T$. The cut identity rewrites "$h(N_T) > 1$ and $\sigma(N_T)$ factors over S" as flow.

- **Lemma 4.9 is classical in substance.** It is equivalent to: if M is a unitary divisor of N with $p \nmid M$, then $\sigma(M)$ has a prime factor not dividing M. The SCC phrasing is pleasant but not new arithmetic. The same is true of our own Proposition 4.11: it is the graph form of the classical fact that a proper unitary divisor M of a perfect number cannot satisfy $M \mid \sigma(M)$,

because $1 < h(M) < 2$. What the graph language adds is the *consequence*—irreducibility, and hence the rank statement of Corollary 4.13—rather than new arithmetic.

- **Lemma 4.14 is weaker than a classical bound.** A primitive prime for $q^{e_q} - 1$ is $\geq 2e_q + 1$ and lies in S, so
$$2e_q + 1 \leq R \qquad \text{for every } q \in Q.$$
In particular $R \nmid e_q$, which already gives $R \nmid e_q / \operatorname{ord}_R(q)$. This applies equally to Rodgers' Theorem 7.2, the source of the lemma, which uses only the weaker bound $\ell \geq e_q + 1$ for a primitive prime ℓ even though, ℓ and e_q both being odd, $\ell \equiv 1 \pmod{2e_q}$ (as the preprint itself notes in its Appendix A.2).

- **The count (7.1) is useless in practice.** The largest prime factor of an OPN exceeds 10^8 (Goto–Ohno [14]), so $h_R - 1$ is vastly larger than any plausible $\omega(N)$.

- **The global inequality in Lemma 4.7 is a tautology.** Since $\sigma_0(e) \leq e$, we have $\sigma_0(e_q) - 1 \leq e_q - 1 = a_q$, hence $\sum_q (\sigma_0(e_q) - 1) \leq \sum_q a_q \leq \Omega(N)$ for *any* exponents whatsoever. This is a warning sign for the Hall framing.

5.3 The main problem: equivalence, not reduction

Claim A. The hypotheses "σ-closure + (L1) + Euler form" are just $\sigma(N) = 2N$ written prime by prime. So the configurations Claim A quantifies over *are* odd perfect numbers, and Claim A says "every OPN violates a necessary condition," i.e. "no OPN exists." A reformulation helps only if one proves deficiency from a *strict subset* of the constraints. But the unit-demand relaxation (H) is almost certainly satisfiable:

- capacities a_r grow linearly in the exponents;
- demands $\sigma_0(e_q) - 1$ grow like a divisor function;
- the tautology above shows the global inequality has built-in slack.

The binding information is *size*: $\Phi_d(q) \approx q^{\varphi(d)}$ must be absorbed *completely* by S with bounded exponents. Restoring it gives a weighted flow with demand $\log \Phi_d(q)$ and capacity $a_r \log r$, which is (L5) plus the full factorization, i.e. the original problem.

Claim B. "OPNs have small exponent excess" contradicts known lower bounds, so it is as hard as nonexistence. The real precedent is the special-exponent literature: Steuerwald [28] ruled out all $\beta_q = 1$, Kanold [19] ruled out all $\beta_q = 2$, and later work by McDaniel, Hagis, Cohen–Williams and others treated further restricted patterns. Even those cases took real effort. The proposed charging scheme needs a recycling bound independent of $\omega(N)$, and nothing in the program supplies one.

A striking recent illustration of how rigid a restricted exponent pattern can be is due to Ochem–Zelinsky [25]. They call an OPN *restrained* when all nonspecial even exponents are 2 except one. In that situation the exceptional component must be a power of 3, and its exponent is at least 23,000,000,000. This does not approach the full nonexistence problem, but it is strong evidence that the "local type + reducible configuration" philosophy in Section 6 can extract serious arithmetic information from nearly minimal exponent patterns.

5.4 Why the obvious spectral normalization is weak

The original Gemini/Argon proposal suggested normalized Laplacians and Cheeger-type expansion. There is a simple reason that the most immediate normalization does not itself create a contradiction. From (L1), the sum of column r of W is a_r. Hence

$$P = WD^{-1}$$

is column-stochastic:

$$\mathbf{1}^{\mathsf{T}} P = \mathbf{1}^{\mathsf{T}}.$$

Since P is nonnegative, its spectral radius is therefore $\rho(P) = 1$, and $I - P$ automatically has eigenvalue 0. By Proposition 4.11, P is moreover irreducible, so by Perron–Frobenius this eigenvalue is simple. Thus the basic Perron eigenvalue is forced by valuation balance and strong connectivity, and contains no new arithmetic information.

A two-vertex caricature. The point is easiest to see in a purely graph-theoretic toy example. The column-stochastic matrix

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

has eigenvalues 1 and -1, while

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

has eigenvalues 1 and 0. Stochasticity forces the eigenvalue 1 but places very different constraints on the rest of the spectrum. Therefore a spectral proof would need an arithmetic theorem explaining why matrices that come from cyclotomic valuations cannot have the "wrong" second eigenvalue; the normalization alone cannot provide that theorem.

One could still study a second eigenvalue, a singular-value gap, or one of the directed normalized Laplacians of Chung [12]. But any useful bound would have to come from the number-theoretic restrictions on the *entries* of W—multiplicative orders, cyclotomic layers, and LTE—not from stochasticity alone. In particular, there is no evident uniform Cheeger gap: the logarithmic divergences satisfy

$$0 < \delta_q < \log \frac{q}{q-1} \sim \frac{1}{q},$$

so large-prime vertices naturally permit cuts that are very nearly balanced in the logarithmic flow. This is why the later discharging and local configuration program seems more concrete than beginning with a normalized Laplacian.

5.5 Sanity filters

Spoofs. Spoof odd perfect numbers exist (Section 1.2). Any argument that does not use the primality of each base may therefore also "prove" that spoofs are impossible, which would be a contradiction. Recall the Descartes spoof $D = 3^2 \cdot 7^2 \cdot 11^2 \cdot 13^2 \cdot 22021$ of Example 1.13.

The log identities (L5)–(L6) hold for the spoof if 22021 is treated as a single vertex: they only use $\prod \Sigma_s = D$. Valuation balance (L1) and the order arguments of Lemmas 4.3–4.7 and 4.14 do distinguish real primes. So these are where primality does real work.

Example 5.1 (A logarithmic 5×5 aggregation of the Descartes spoof)**.** The Descartes spoof $D = 3^2 \cdot 7^2 \cdot 11^2 \cdot 13^2 \cdot 22021$ from Example 1.13 satisfies $22021 = 19^2 \cdot 61$. Treating $C = 22021$ as a single pseudo-prime cannot produce an honest valuation matrix, because v_C is not a prime-adic valuation. It does, however, produce an exact *logarithmic* aggregation.

Put

$$\theta = \frac{\log 19}{\log C}, \qquad \psi = \frac{\log 61}{\log C}.$$

Since $C = 19^2 \cdot 61$,

$$2\theta + \psi = 1.$$

The identities

$$\begin{aligned} \Sigma_3 &= 13, & \Sigma_7 &= 3 \cdot 19 = 3C^\theta, \\ \Sigma_{11} &= 7 \cdot 19 = 7C^\theta, & \Sigma_{13} &= 3 \cdot 61 = 3C^\psi, \\ \Sigma_C &= (C+1)/2 = 7 \cdot 11^2 \cdot 13 \end{aligned}$$

therefore yield, in the coordinate order $(3, 7, 11, 13, C)$, the real matrix

$$L_{\text{eff}} = \begin{pmatrix} 2 & 0 & 0 & -1 & 0 \\ -1 & 2 & 0 & 0 & -\theta \\ 0 & -1 & 2 & 0 & -\theta \\ -1 & 0 & 0 & 2 & -\psi \\ 0 & -1 & -2 & -1 & 1 \end{pmatrix}.$$

Its columns sum to zero precisely because $2\theta+\psi = 1$, and with $\ell = (\log 3, \log 7, \log 11, \log 13, \log C)^{\mathsf{T}}$ it satisfies the same formal Poisson equation $L_{\text{eff}}\ell = -\delta$ for the five spoof components.

The point is diagnostic: L_{eff} has *real logarithmic coefficients*, not prime-adic valuation weights. The earlier values $1/2, 1/2$ would not reproduce $\log 19$ and $\log 61$ and hence do not give the correct Poisson equation. Splitting C into its true prime factors 19 and 61 also does not create a genuine six-prime σ-graph for the spoof, because the outgoing pseudo-prime local sum $(C+1)/2$ is not $\sigma(19^2)\sigma(61)$. This is exactly the arithmetic information lost when a composite is allowed to masquerade as a prime.

Relaxation feasibility. Before trying to prove that a class of configurations is infeasible, check by computer that the *relaxed* constraints really are infeasible. A useful workflow is to choose a small candidate prime set S, prescribe or bound the exponents a_s, introduce integer variables $w(s, r)$, impose the exact column balances (L1), and forbid edges that violate the order conditions of Lemma 4.3. One may then solve the resulting integer or linear feasibility problem without requiring every row to be the complete prime factorization of an actual $\sigma(s^{a_s})$.

If this relaxed system already has no solution, the solver can often expose a small dual or combinatorial certificate worth proving by hand. If it *does* have solutions, that is equally informative: any proposed proof using only those relaxed constraints cannot work, because the fake solution exhibits exactly what arithmetic information has been discarded. In this sense computation is a sanity filter for the proof strategy, not evidence for or against the existence of an actual OPN.

6 A stronger graph-theoretic approach

No known graph-theoretic method currently approaches a proof of nonexistence. The realistic target is instead *structural inequalities*: bounds on how many primes can have small exponent,

ratio bounds on Ω/ω, restrictions on short σ-cycles, and the like. Graph language already has a track record here: the weighted digraph of Bibby–Vyncke–Zelinsky [7] is used to organize $\sigma_{m,n}$-pair restrictions, while modern factor-chain/discharging arguments prove quantitative bounds on prime factors.

6.1 Discharging

Ochem–Rao's inequalities $\Omega \geq (18\omega - 31)/7$ and $\Omega \geq 2\omega + 51$ [23] are discharging arguments in disguise. (Ochem is a graph theorist.) One can make this explicit.

Zelinsky subsequently improved the asymptotic slope: if $3 \mid N$ then

$$\Omega(N) \geq \frac{66}{25}\omega(N) - 5,$$

while if $3 \nmid N$ then

$$\Omega(N) \geq \frac{302}{113}\omega(N) - \frac{286}{113}$$

[31]. Clayton and Hansen [10] then improved the slope again: for every OPN

$$\Omega(N) \geq \frac{99}{37}\,\omega(N) - \frac{187}{37} \qquad (99/37 \approx 2.676),$$

and if $3 \nmid N$,

$$\Omega(N) \geq \frac{51}{19}\,\omega(N) - \frac{46}{19} \qquad (51/19 \approx 2.684).$$

Ochem's project page currently summarizes the state of the art as $\Omega(N) \geq \max\big(115,\ 2\omega(N) + 51,\ (99\omega(N) - 187)/37\big)$, the 115 being an unpublished computational bound [24]. These inequalities are useful benchmarks for any LP/discharging model derived directly from G_N.

- **Charges.** Give $s \in S$ initial charge $c(s) = a_s - 2$, so a low-exponent vertex ($a_q = 2$) starts at 0. A ratio bound $\Omega \geq \lambda\omega$ with $\lambda > 2$ says low-exponent vertices cannot dominate.
- **Rules.** Move charge along edges between low- and high-exponent vertices, and show every vertex ends with charge $\geq \varepsilon$.
- **Rigid local structure.** If $a_q = 2$ then $\Sigma_q = \Phi_3(q) = q^2 + q + 1$. If $q \equiv 1 \pmod 3$ then $q^2 + q + 1 \equiv 3 \pmod 9$, so $v_3 = 1$; every other prime factor r has $\mathrm{ord}_r(q) = 3$, hence $r \equiv 1 \pmod 6$. The whole of $q^2 + q + 1$ must be absorbed by S.
- **Reducible configurations become Diophantine equations.** Patterns such as $\Phi_3(q) = 3r$, $\Phi_3(q) = r^k$, or a short cycle of $a = 2$ vertices exchanging all their valuation reduce to Nagell–Ljunggren-type equations $\frac{x^n-1}{x-1} = y^m$ or to S-unit problems. Three classical positive solutions of the former are

  $$\tfrac{3^5-1}{2} = 11^2, \qquad \tfrac{7^4-1}{6} = 20^2, \qquad 18^2 + 18 + 1 = 7^3.$$

  The general Nagell–Ljunggren equation is *not* completely solved; Bennett–Levin [6] emphasize that proving completeness of the known list is beyond current methods, while obtaining strong restrictions on any further solution. Thus a discharging proof may use classified small subcases, but should not silently assume a complete solution of the general equation.

What discharging is doing here. A discharging argument is bookkeeping: total charge is conserved while it is moved between vertices according to local rules. With the simple initial choice $c(s) = a_s - 2$, a nonspecial exponent-2 vertex has charge 0, an exponent-4 vertex has charge 2, and an Euler vertex with $\alpha = 1$ has charge -1. The total initial charge is

$$\sum_s c(s) = \Omega(N) - 2\omega(N).$$

If arithmetic restrictions force high-exponent vertices to subsidize nearby low-exponent configurations in such a way that every vertex finishes with a positive average charge, then one obtains a lower bound on Ω/ω.

A toy model makes the mechanism visible. Ignore the special vertex and suppose every vertex has exponent either 2 or at least 4, and suppose one could prove the local structural inequality

$$x_2 \le 2x_{\ge 4},$$

where x_2 and $x_{\ge 4}$ count the two types. Then, using the smallest possible exponent in each class,

$$\frac{\Omega}{\omega} \ge \frac{2x_2 + 4x_{\ge 4}}{x_2 + x_{\ge 4}} \ge \frac{8}{3}.$$

The inequality $x_2 \le 2x_{\ge 4}$ is hypothetical here; the point is that a local graph restriction immediately turns into a global prime-factor bound. A dual LP or a discharging proof is a systematic way to discover and certify such inequalities.

This is the architecture of the Four Color Theorem proofs: an unavoidable set of local configurations, each shown reducible, with discharging connecting them. In the OPN setting the "reducibility" step is number-theoretic rather than planar-topological: one rules out a local σ-graph pattern by congruences, multiplicative orders, cyclotomic factorization, or a Diophantine equation.

6.2 LP duality finds the rules

1. Classify vertices into *types*: exponent class ($a = 2, 4, 6, \ldots$, or special), small-prime residue data ($q \bmod 3$, $q \bmod 5$, ...), and whether $q = R$.

2. Write (L1) and (L5) as linear constraints in the variables x_t (number of vertices of type t) and $y_{t,t'}$ (total weight from type t to type t'). Lemma 4.3 decides which $y_{t,t'}$ may be positive and with what minimum.

3. Minimize $\Omega/\omega = \sum_t a_t x_t / \sum_t x_t$ over the LP relaxation. The optimal dual solution *is* a set of discharging rules.

4. The extremal primal solution shows which configurations are "too cheap"; kill them with Diophantine or Zsigmondy arguments, add the exclusions as constraints, and re-solve.

Each round is a certified improvement, checkable independently.

Why the dual solution is useful. The primal LP asks, in effect, "what is the cheapest hypothetical mixture of vertex types that obeys all currently known local rules?" If the optimum is too low, its extremal solution identifies the configuration responsible for the weakness. The dual LP assigns numerical prices to the constraints. A dual certificate can often be rewritten as a human discharging proof: the "prices" say how much charge each local rule is allowed to transfer.

For example, in the toy inequality $x_2 \le 2x_{\ge 4}$ above, one may think of a high-exponent vertex as having enough surplus to support at most two low-exponent vertices. The resulting $8/3$ bound is what a tiny two-type LP would discover automatically. A realistic OPN LP would have many more types (residue classes, exponent classes, maximal-prime status, and perhaps cyclotomic-layer data), but the logic is the same: solve the relaxation, inspect the cheapest pattern, prove that pattern arithmetically impossible, and add the corresponding inequality.

6.3 Flow decomposition

Weight edge $s \to r$ by $w(s,r)\log r$. Each $q \ne p$ has in-flow $a_q \log q$ and out-flow $a_q \log q + \delta_q$, with $0 < \delta_q < \log \frac{q}{q-1}$. Thus the flow is a near-circulation *in divergence*: there are small sources at every $q \ne p$ and one sink at p. The standard flow-decomposition theorem writes a nonnegative flow as directed cycles plus source-to-sink paths, and the total amount *injected* into the path part is $\sum_{q \ne p} \delta_q = -\delta_p$. Small injected flow does not immediately mean small path *mass*, because a little flow could in principle travel along many edges. But the paths in a flow decomposition can be taken to be simple, and a simple path in a graph with $\omega(N)$ vertices has at most $\omega(N) - 1$ edges. That is enough to make the slogan "most of the mass circulates" precise.

Proposition 6.1 (Most logarithmic mass lies on cycles)**.** *Decompose the flow $f(s,r) = w(s,r)\log r$ into simple directed cycles and simple paths from the sources $q \ne p$ to the sink p. The total edge mass $\sum_{s,r} f(s,r)$ equals $\log N$, while the edge mass carried by the paths is at most*

$$(\omega(N) - 1)\log \frac{2p^\alpha}{\sigma(p^\alpha)} < (\omega(N) - 1)\log 2.$$

Consequently the fraction of the edge mass carried by cycles exceeds

$$1 - \frac{(\omega - 1)\log 2}{\log 5 + 2(\omega - 1)\log 3} \;>\; 1 - \frac{\log 2}{2\log 3} \;>\; 0.68,$$

and this fraction tends to 1 as $\omega(N) \to \infty$.

Proof. By (L1), $\sum_{s,r} w(s,r)\log r = \sum_r a_r \log r = \log N$. The path flows add up to the total supply $\sum_{q \ne p} \delta_q = -\delta_p = \log\bigl(2p^\alpha/\sigma(p^\alpha)\bigr)$, which is less than $\log 2$, and each path uses at most $\omega - 1$ edges. For the denominator, $\log N = \sum_s a_s \log s \ge \log 5 + 2\sum_{q \in Q} \log q$, and the $\omega - 1$ primes $q \in Q$ are distinct odd primes, so $\sum_{q \in Q} \log q \ge (\omega - 1)\log 3$; in fact this sum grows like $\omega \log \omega$, which gives the limit. □

For instance, with $\omega(N) = 10$ the paths carry less than $9\log 2/(\log 5 + 18\log 3) \approx 29\%$ of the logarithmic mass, so more than 70% of it travels around directed cycles. This is a structural statement only, but it justifies paying special attention to cycles:

- *Short cycles are rigid.* The 2-cycle of Example 4.5 exists, but a cycle carrying most of the valuation at both ends forces near-equalities $q^2 + q + 1 = 3^\varepsilon r^k \cdot (\text{small})$, which fall under S-unit bounds.
- *Long cycles force growth.* Primitive edges satisfy $\ell \equiv 1 \pmod{2d}$, and the maximal prime caps $2e_q + 1 \le R$.

6.4 A concrete first project

A useful first project should have an external benchmark: one should know in advance what inequality the graph machinery ought to reproduce before asking it to improve anything. The following sequence provides such a test.

1. **Re-derive a known bound in graph language.** Recast $\Omega \geq (18\omega - 31)/7$ as an explicit discharging proof on G_N. The goal is not novelty; it is to verify that the chosen vertex types and local edge rules are rich enough to recover established arithmetic.

2. **Let an LP rediscover the same rules.** Build the type-LP in Sage and compare its optimum with 18/7. If the LP gives a weaker slope, inspect its extremal solution to see exactly which arithmetic condition is missing. If its dual reproduces the known discharging coefficients, that is evidence that the modeling choices are faithful.

3. **Add one genuinely arithmetic exclusion.** Encode classified Nagell–Ljunggren or S-unit exclusions for selected $a = 2$ configurations and re-solve. Any improvement in the optimum is then traceable to a specific theorem rather than to an opaque numerical search.

Zelinsky's 2018 paper [30] and the sharper 2021 result [31] improve Ochem–Rao's 18/7 slope by analyzing local component/factor structures, and Clayton–Hansen [10] improve them further. Any implementation should benchmark against the current record slopes 99/37 (general) and 51/19 (when $3 \nmid N$), as well as the computational bounds reported by Ochem [24].

6.5 Arithmetical structures on the σ-graph

Corollary 4.13 says that the Poisson operator $L = D - W$ of an OPN has a one-dimensional kernel and that its left kernel is spanned by $\mathbf{1}$. This subsection identifies the kernel. The right language is Lorenzini's theory of arithmetical structures [20], recalled in Section 2.5. The upshot is that an OPN σ-graph carries two arithmetical structures whose S-values are both the exponent vector a: a trivial one on incoming edges, and a canonical, nontrivial one on outgoing edges.

6.5.1 The two structures on an OPN σ-graph

Proposition 6.2 (The in-structure is natural)**.** *For an OPN, $(R, S) = (\mathbf{1}, a)$ is an in-structure on G_N, with generalized Laplacian $L = D - W$.*

Proof. With $R \equiv 1$ the defining identity reads $a_r = \sum_s w(s, r)$, which is valuation balance (Lemma 4.2). □

Since a_r is the weighted in-degree of r, this is the directed analogue of the natural structure $R \equiv 1$, $S = \deg$. It contains no new information; it is (L1) in different clothing. The out-direction is more interesting.

Theorem 6.3 (The canonical out-structure)**.** *Let N be an OPN. There is a unique positive integer vector $R_N = (R_N(s))_{s \in S}$ with $\gcd R_N = 1$ such that (R_N, a) is an out-structure on G_N:*

$$a_s \, R_N(s) = \sum_{r \in S} w(s, r) \, R_N(r) \qquad (s \in S).$$

Equivalently, R_N spans $\ker L$.

Proof. The matrix $D^{-1}W = D^{-1}(WD^{-1})D$ is similar to WD^{-1}, which is column-stochastic by (L1), so its spectral radius is 1. By Proposition 4.11 the digraph G_N is strongly connected, so $D^{-1}W$ is irreducible. The Perron–Frobenius theorem (compare the undirected version in the subsection "Perron–Frobenius" of the chapter "Applications to graph spectra" of [18]) gives a positive vector x with $D^{-1}Wx = x$, i.e. $Lx = 0$, unique up to scaling. Because L has integer entries and rank $\omega(N)-1$ (Corollary 4.13), its kernel is spanned by a rational vector; scaling x to a primitive integer vector gives R_N. □

So the single integer matrix $L = \operatorname{diag}(a) - W$ is simultaneously the generalized Laplacian of a trivial in-structure (left kernel $\mathbf{1}$) and of a nontrivial out-structure (right kernel R_N). In the symmetric theory of [18] these two kernels coincide; here they need not.

Corollary 6.4 (Properties of R_N)**.** *Let N be an OPN.*

(a) $R_N \equiv \mathbf{1}$ *if and only if* $\Omega(\Sigma_s) = a_s$ *for every* $s \in S$*, i.e. if and only if every vertex of* G_N *has equal weighted in- and out-degree.*

(b) The solutions of $Lx = -\delta$ *are exactly the vectors* $\ell + tR_N$*,* $t \in \mathbb{R}$*, where* $\ell = (\log s)_{s\in S}$*.*

(c) $\operatorname{coker} L \cong \mathbb{Z} \oplus K(N)$ *for a finite abelian group* $K(N)$*, the arithmetical critical group of the out-structure.*

Proof. (a) $L\mathbf{1} = (a_s - \Omega(\Sigma_s))_s$, so $\mathbf{1} \in \ker L$ exactly when every $a_s = \Omega(\Sigma_s)$; by uniqueness, this happens exactly when $R_N = \mathbf{1}$. (b) Corollary 4.13. (c) L is an integer matrix of rank $\omega(N) - 1$; its Smith normal form has exactly one zero on the diagonal. □

Part (a) gives a pleasant interpretation: R_N measures how far the σ-graph is from being *Eulerian* (balanced), which is the same defect $V_s = a_s - \Omega(\Sigma_s)$ that appears as the "potential" in the quadratic-form identity of Section 6.6. Part (b) says that R_N is exactly the ambiguity in the Poisson equation. Part (c) attaches to every hypothetical OPN a finite abelian group $K(N)$, about which nothing is currently known.

6.5.2 Examples from multiperfect numbers

The same construction works for k-perfect numbers, using the vertex set $\widehat{S}$ of Section 2.2 and S-values $a_s + v_s(k)$ (the column sums of W). In the examples below every prime of k divides N, the digraph is strongly connected, and the kernel of $\operatorname{diag}(a+v(k)) - W$ is one-dimensional and spanned by a positive vector; all of this was checked by computer.

Example 6.5 (Even perfect numbers)**.** For $N = 2^{p-1}M_p$ the σ-graph is the 2-cycle with $w(2, M_p) = 1$ and $w(M_p, 2) = p$, and the S-values are $(p, 1)$. The out-structure is

$$R = (1, p), \qquad S = (p, 1),$$

since $p \cdot 1 = w(2, M_p) \cdot p$ and $1 \cdot p = w(M_p, 2) \cdot 1$. The generalized Laplacian $\begin{pmatrix} p & -1 \\ -p & 1 \end{pmatrix}$ has Smith normal form $\operatorname{diag}(1, 0)$, so the critical group is trivial. By Corollary 6.4(a), $R \neq \mathbf{1}$ reflects the imbalance between the in-degree p and out-degree 1 at the vertex 2.

Example 6.6 ($N = 120$ and the triangle)**.** For $120 = 2^3 \cdot 3 \cdot 5$ (Example 2.12), in the order $(2, 3, 5)$,

$$W = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \qquad S = a + v(3) = (3, 2, 1).$$

Solving $(\mathrm{diag}(S) - W)R = 0$ gives $R = (1, 1, 2)$:

$$3 \cdot 1 = 1 \cdot 1 + 1 \cdot 2, \qquad 2 \cdot 1 = 2 \cdot 1, \qquad 1 \cdot 2 = 1 \cdot 1 + 1 \cdot 1.$$

The underlying graph of this σ-graph is the triangle C_3, and R is the same vector $(1, 1, 2)$ as in the arithmetical structure on C_3 in [18]. Only the S-values differ, $(3, 2, 1)$ instead of $(3, 3, 1)$, because the edges now have directions and unequal weights. The Smith normal form of the generalized Laplacian is $\mathrm{diag}(1, 2, 0)$, so $K(120) \cong \mathbb{Z}/2$.

N	k	primes	R	S
$2^{p-1}M_p$	2	$2, M_p$	$(1, p)$	$(p, 1)$
120	3	$2, 3, 5$	$(1, 1, 2)$	$(3, 2, 1)$
672	3	$2, 3, 7$	$(1, 1, 3)$	$(5, 2, 1)$
30240	4	$2, 3, 5, 7$	$(1, 2, 3, 3)$	$(7, 3, 1, 1)$
32760	4	$2, 3, 5, 7, 13$	$(1, 2, 3, 3, 4)$	$(5, 2, 1, 1, 1)$
2178540	4	$2, 3, 5, 7, 13, 19$	$(2, 5, 7, 8, 10, 11)$	$(4, 2, 1, 2, 1, 1)$

The critical groups are $\mathbb{Z}/2$ for 120, 672, 30240 and 32760, and trivial for the even perfect numbers and for 2178540.

6.5.3 Equitable quotients force R_N to be constant on cells

The pullback construction of [18] transports an arithmetical structure from the target of a harmonic morphism to its source, with R constant on fibers. Because R_N is *unique*, this can be run backwards to give a concrete obstruction to equitable quotients of G_N.

Call a partition π of S *out-equitable* if for any two cells C, C' the row sum $\sum_{r \in C'} w(s, r)$ takes the same value $b_{CC'}$ for every $s \in C$. This is the directed form of the weighted equitable partitions of the chapter "Edge-weighted graph partitions" of [18]: each vertex of a cell sends the same total weight to each cell. (It is the transpose of the in-equitable condition of Section 2.4.)

Proposition 6.7 (Descent of the canonical structure)**.** *Let N be an OPN and let π be an out-equitable partition of S such that the exponents a_s are constant, say a_C, on each cell C. Then R_N is constant on each cell, and $(R_N|_C, a_C)$ is an out-structure on the quotient digraph with weight matrix $B = (b_{CC'})$, where the diagonal entries b_{CC} are recorded as loops.*

Proof. Let U be the space of vectors constant on cells. Out-equitability says $WU \subseteq U$, and $DU \subseteq U$ because a is constant on cells, so $LU \subseteq U$. Since $\mathbf{1}^{\mathsf{T}} L = 0$, the image LU lies in the proper subspace $U \cap \{x : \mathbf{1}^{\mathsf{T}} x = 0\}$ of U, so $L|_U$ is not surjective and hence has a nonzero kernel. That kernel lies in $\ker L = \mathbb{R}R_N$, so $R_N \in U$. For $s \in C$, $a_C R_N(C) = \sum_{C'} \sum_{r \in C'} w(s, r) R_N(C') = \sum_{C'} b_{CC'} R_N(C')$, which is the out-structure identity on the quotient. □

Corollary 6.8 (An equitability test)**.** *If two primes of an OPN have the same exponent but different values of R_N, then no out-equitable partition with exponents constant on cells can place them in the same cell.*

Example 6.9. The same argument applies to the multiperfect examples above. For $N = 2178540 = 2^2 \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 13 \cdot 19$, the primes 3 and 7 both have S-value 2 but R-values 5 and 8, and 5, 13, 19 all have S-value 1 but R-values 7, 10, 11. Since all six R-values are distinct, the only out-equitable partition with S constant on cells is the partition into singletons. This gives a concrete, checkable form of the expectation, stated in Section 7, that no natural partition of an OPN σ-graph is equitable.

6.5.4 Undirected analogues

The undirected models of Section 6.6 carry only the natural structure.

- **σ-symmetric graphs.** If $W = W^{\mathsf{T}}$, then (L1) makes the degree of s equal to a_s, so the in- and out-structures coincide with the natural structure $R \equiv 1$, $S = a = \deg$. By the handshake theorem (Theorem 6.16) no OPN arises this way. For the σ-closed configuration $3^3 \cdot 5$ of Example 6.15, the natural structure has $S = (1, 1)$ while the exponents are $(3, 1)$; the mismatch is exactly the failure of valuation balance.
- **The source–target bipartite graph B_N.** Its natural structure has $S(r^+) = a_r$ on the target side (this is (L1) again) and $S(s^-) = \Omega(\Sigma_s)$ on the source side. The canonical R_N does not lift: for $N = 120$, putting $R = (1, 1, 2)$ on both copies would force $S(5^+) = 1/2$.
- **The symmetrization $W + W^{\mathsf{T}}$.** The same test fails for $N = 120$, where it would force $S(5) = 3/2$, and no canonical structure comes from the OPN data.

The genuinely new arithmetical structure is therefore the directed one. It is canonical, it carries a critical group $K(N)$, and its R-values give a working obstruction to equitable quotients.

6.6 Undirected avatars of the σ-graph

The σ-graph is naturally directed because the arithmetic question "which local divisor sum supplied this prime?" has a source and a target. Nevertheless, several useful undirected formulations exist. They should not all be confused with the naive operation of simply erasing arrowheads. Some are *exact encodings* of the directed data; others deliberately forget information in exchange for access to matching theory, ordinary cut theory, or symmetric spectral methods.

6.6.1 The source–target bipartite graph: an exact undirected encoding

The most faithful undirected construction is the source–target bipartite graph of Section 2.6. Write $B_N = B(G_N)$. Think of s^- as the *supplier copy* of the prime s and r^+ as the *receiver copy* of r. The biadjacency matrix of B_N is W, so nothing is lost as long as the two color classes and the prime labels are kept.

Example 6.10 (What one row looks like)**.** Take the familiar local calculation

$$\sigma(13^2) = 183 = 3 \cdot 61.$$

The directed graph has edges $13 \to 3$ and $13 \to 61$, both of weight 1. In B_N these become

$$13^- \text{ --- } 3^+, \qquad 13^- \text{ --- } 61^+.$$

Similarly,

$$\sigma(3^4) = 121 = 11^2$$

produces one weighted edge $3^- = = 11^+$ of weight 2; if one prefers an unweighted multigraph, this can be pictured as two parallel edges. Thus a weighted source vertex records the complete prime factorization of its local divisor sum.

The perfectness equation becomes especially transparent on the target side. Lemma 4.2 says

$$\boxed{\deg_{B_N}(r^+) = a_r.}$$

Here degree means *weighted* degree. In words: the receiving copy of r must receive exactly as many units of r-adic valuation as the exponent of r in N. The source degree has a different meaning:

$$\deg_{B_N}(s^-) = \sum_r w(s,r) = \Omega(\Sigma_s),$$

the number of prime factors of the odd local divisor sum, counted with multiplicity. The number theory is therefore asymmetric even though the new graph is undirected: the asymmetry has moved from the arrows into the two color classes.

This is an important conceptual point. Directedness is not essential to *encoding* the problem. What is essential is remembering which copy of a prime is acting as a supplier and which is acting as a receiver.

6.6.2 Cyclotomic obligations as an undirected b-matching problem

Claim A in the breakthrough section has an even more natural undirected form. Make a bipartite graph with the cyclotomic obligations

$$\mathcal{O} = \{(q,d) : q \in Q,\ d \mid e_q,\ d > 1\}$$

on the left and the prime set S on the right. Join (q,d) to r when

$$\mathrm{ord}_r(q) = d.$$

Thus an edge means that r is arithmetically eligible to serve as a primitive prime divisor for the layer $\Phi_d(q)$. Give every obligation capacity 1 and every prime vertex capacity a_r. A feasible assignment of obligations to targets is then a b-matching.

For a set $\mathcal{A}$ of obligation vertices, let $N(\mathcal{A})$ denote its neighborhood of eligible prime targets. The capacitated Hall condition is

$$\boxed{|\mathcal{A}| \le \sum_{r \in N(\mathcal{A})} a_r.}$$

This is precisely the combinatorial inequality appearing in Claim A.

Example 6.11 (One obligation versus valuation multiplicity)**.** For 13^2, the primitive obligation is $(13,3)$. Since $\mathrm{ord}_{61}(13) = 3$, the bipartite obligation graph contains

$$(13,3) \ - - - \ 61.$$

The factor 3 in $\Phi_3(13) = 3 \cdot 61$ is *not* a primitive target for this obligation, because $\mathrm{ord}_3(13) = 1$; it belongs to the separate order-one mechanism of Lemma 4.3.

Now compare 3^4, where

$$\Phi_5(3) = 11^2.$$

There is only one primitive obligation, $(3,5)$, but satisfying it consumes two units of 11-adic valuation. A unit-demand Hall model records merely that $(3,5)$ can be assigned to 11; the full valuation graph remembers the weight 2. This is why the b-matching picture is useful but is a *relaxation*, not a replacement for the complete factorization problem.

The advantage of this reformulation is methodological. Matching theory can identify a small set of obligations that competes for too little target capacity. If such a Hall-deficient set could be forced from only a subset of the OPN constraints, it would give a genuinely local obstruction rather than merely restating perfectness.

6.6.3 An undirected current and Kirchhoff's law

There is also an exact undirected version of the logarithmic cut identity. Form the ordinary undirected support graph on S: put an edge $\{s, r\}$ when at least one of $w(s, r)$ and $w(r, s)$ is nonzero. On each such undirected edge define the antisymmetric current

$$\boxed{J_{sr} = w(s, r) \log r - w(r, s) \log s, \qquad J_{rs} = -J_{sr}.}$$

Choosing the orientation $s \to r$ is only a sign convention; if $J_{sr} < 0$, the net current actually points the other way.

Its divergence at s is

$$\begin{aligned} \operatorname{div} J(s) &= \sum_r J_{sr} \\ &= \sum_r w(s, r) \log r - \log s \sum_r w(r, s) \\ &= \log \Sigma_s - a_s \log s \\ &= \delta_s, \end{aligned}$$

where the second sum is a_s by exact valuation balance. Hence

$$\boxed{\operatorname{div} J = \delta.}$$

This is Kirchhoff's current law with a source of strength $\delta_q > 0$ at each nonspecial prime and a single sink of strength $-\delta_p$ at the Euler prime. Global conservation

$$\sum_s \delta_s = 0$$

means that the total injected current equals the current absorbed at p.

Summing over a vertex set T gives the undirected cut law

$$\boxed{\sum_{s \in T} \delta_s = \sum_{\substack{s \in T \\ r \notin T}} J_{sr}.}$$

Thus Lemma 4.8 can be read in the familiar language "net current through the boundary equals total source strength inside." In particular, if T contains no Euler prime, its total source strength is positive, so the net logarithmic current crosses the cut outward.

Example 6.12 (A two-way arithmetic edge)**.** In the primitive 2-cycle of Example 4.5, both $w(13, 61)$ and $w(61, 13)$ equal 1. If we orient the underlying undirected edge from 13 to 61, then

$$J_{13,61} = \log 61 - \log 13 > 0.$$

The formula has compressed the two opposite directed edges into one signed number without forgetting their competition. A one-way edge would simply contribute $w(s, r) \log r$ in its forward direction.

6.6.4 Why this is not an ordinary resistor network

The Kirchhoff analogy is exact at the level of divergence, but one should not silently import all of electrical-network theory. In an ordinary resistor network, current is generated from a potential V by Ohm's law

$$J_{sr} = c_{sr}(V_s - V_r), \qquad c_{sr} \geq 0,$$

where c_{sr} is the edge conductance. The natural arithmetic potential is $V_s = \log s$, but the OPN current need not have the sign of $\log s - \log r$.

The $13 \leftrightarrow 61$ example makes the obstruction concrete. We found

$$J_{13,61} = \log 61 - \log 13,$$

whereas

$$V_{13} - V_{61} = \log 13 - \log 61.$$

Therefore an Ohm-law representation on this edge would require

$$c_{13,61} = -1.$$

So the exact arithmetic current naturally behaves like a *signed* network, not necessarily a positive-conductance electrical network. This is not merely a nuisance: it identifies precisely why standard undirected potential theory does not immediately solve the OPN problem.

The sign is not an accident of this example. On any edge where the two directed weights agree, say $w(s, r) = w(r, s) = c$, the current is

$$J_{sr} = c\,(\log r - \log s) = -c\,(V_s - V_r),$$

so Ohm's law holds with conductance exactly $-c$: current always flows *uphill*, toward the larger prime. Section 6.6.7 below studies what happens when *every* edge is of this kind.

6.6.5 A symmetric spectral avatar

The source–target bipartite graph also gives a canonical way to replace the nonsymmetric matrix W by a symmetric matrix. Define

$$\boxed{\mathcal{A}_W = \begin{pmatrix} 0 & W \\ W^{\mathsf{T}} & 0 \end{pmatrix}.}$$

This is exactly the weighted adjacency matrix of B_N. Because it is real and symmetric, all of its eigenvalues are real. Moreover its nonzero spectrum is

$$\pm\sigma_1(W),\ \pm\sigma_2(W), \ldots,$$

where the $\sigma_i(W)$ are the singular values of W. Thus the most canonical "undirected spectrum" attached to the valuation digraph is the singular-value spectrum of W.

Example 6.13 (A bare two-cycle)**.** If one isolates a two-vertex pattern with

$$W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

then both singular values of W are 1, and the doubled symmetric matrix has spectrum $\{1, 1, -1, -1\}$. The example is intentionally simple: it shows that symmetrization is an exact matrix construction, but it also warns that the spectrum by itself does not remember which arithmetic reason created each entry of W.

One can do the same with the Poisson operator $L = D - W$. Its symmetric dilation is

$$\boxed{\mathcal{L}_{\mathrm{sym}} = \begin{pmatrix} 0 & L \\ L^{\mathsf{T}} & 0 \end{pmatrix}.}$$

Since $L\ell = -\delta$,

$$\mathcal{L}_{\mathrm{sym}} \begin{pmatrix} 0 \\ \ell \end{pmatrix} = \begin{pmatrix} -\delta \\ 0 \end{pmatrix}.$$

So the directed Poisson equation embeds exactly into a symmetric linear system on a doubled vertex set. However, $L = D - W$ contains positive diagonal terms and negative off-diagonal terms, so $\mathcal{L}_{\mathrm{sym}}$ is best regarded as a signed weighted bipartite operator rather than the ordinary Laplacian of an unsigned graph. Its eigenvalues are $\pm$ the singular values of L.

This suggests a better-posed spectral research question than asking for an unspecified directed Cheeger gap:

> Can the cyclotomic, order, and valuation restrictions on an OPN force a singular-value pattern for W or L that no finite source–target bipartite graph with the required degree data can realize?

At present this is a research question, not a known obstruction. The virtue of the formulation is that the symmetric operator is canonical and no arbitrary directed-Laplacian convention has been chosen.

6.6.6 What survives if one simply erases the arrows?

Finally, let U_N be the simple undirected graph on S with an edge $\{q, r\}$ whenever either $q \to r$ or $r \to q$ occurs in G_N. This graph retains adjacency but forgets both direction and valuation. Lemma 4.9 implies that every prime has a directed path to the Euler prime p; therefore

$$\boxed{U_N \text{ is connected.}}$$

(In fact Proposition 4.11 gives the much stronger directed statement that G_N itself is strongly connected.) Connectivity of U_N is a genuine necessary condition, but it is very weak.

For example, the three directed patterns

$$13 \to 61, \qquad 61 \to 13, \qquad 13 \leftrightarrow 61$$

all collapse to the same undirected edge $\{13, 61\}$. The last pattern is a primitive 2-cycle, while the first two are not cycles at all. Therefore the plain underlying graph cannot encode any OPN equivalence by itself.

6.6.7 Genuinely undirected σ-graphs and the handshake lemma

The constructions above all keep the asymmetry of the arithmetic somewhere: in a bipartition, in the sign of a current, or in a block matrix. One can also ask the bolder question: *what if the arithmetic itself were symmetric?* This gives an honest undirected analogue of the OPN problem, and, unlike the original, it can be settled completely.

Definition 6.14 (σ-symmetric configurations)**.** A set of odd primes S with exponents a_s is *σ-symmetric* if

$$w(s,r) = w(r,s) \qquad \text{for all } s,r \in S,$$

i.e. $v_r(\Sigma_s) = v_s(\Sigma_r)$: the prime r occurs in the local divisor sum at s exactly as often as s occurs in the local divisor sum at r. In that case W is the adjacency matrix of an undirected multigraph Γ on S, with $w(s,r)$ parallel edges joining s and r.

The notion is not vacuous.

Example 6.15 (Small σ-symmetric configurations)**.** (a) For $3^2 \cdot 13^2$: $\sigma(3^2) = 13$ and $\sigma(13^2) = 3 \cdot 61$, so $w(3,13) = w(13,3) = 1$. Restricted to $\{3,13\}$ the graph is a single undirected edge 3—13. (The prime 61 leaks out, so this configuration is not σ-closed.)

(b) For $13^2 \cdot 61^2$: the primitive 2-cycle of Example 4.5 has $w(13,61) = w(61,13) = 1$, again a single undirected edge, again not σ-closed (3 and 97 leak out).

(c) A σ-closed example: $N_0 = 3^3 \cdot 5 = 135$. Here

$$\sigma(3^3) = 40 = 2^3 \cdot 5, \qquad \sigma(5) = 6 = 2 \cdot 3,$$

so on odd primes the graph is the single undirected edge 3—5, and every odd prime of $\sigma(N_0) = 2^4 \cdot 3 \cdot 5$ lies in $\{3,5\}$. What fails is valuation balance: the vertex 3 receives one copy of 3, while its exponent is 3. Correspondingly $h(N_0) = 16/9$, not 2.

A small computer search over σ-closed configurations on at most four primes below 400, with exponents at most 6, found that every symmetric one with at least one edge contains exactly this edge 3—5 (together, possibly, with isolated vertices such as 7, 31, 127, whose divisor sums $\sigma(7) = 8$, $\sigma(31) = 32$, $\sigma(127) = 128$ have no odd part).

So symmetric local arithmetic does occur. The "undirected OPN conjecture" is the statement that it can never occur together with perfectness, and that statement is a short exercise in elementary graph theory.

Theorem 6.16 (Handshake obstruction)**.** *No odd perfect number is σ-symmetric.*

Proof. If W is symmetric, then by (L1) the degree of s in the multigraph Γ is

$$\deg_\Gamma(s) = \sum_r w(r,s) = a_s.$$

By the handshake lemma, $\sum_s \deg_\Gamma(s) = 2|E(\Gamma)|$ is even, i.e. Γ has an even number of vertices of odd degree. But Euler's form says that $a_p = \alpha$ is odd and every other $a_q = 2\beta_q$ is even: exactly one vertex of odd degree. Contradiction. □

It is instructive to see why the *directed* graph escapes this argument. In a digraph, the sum of the in-degrees equals the sum of the out-degrees (both are $\Omega(N)$), but that equality carries no parity information: the odd in-degree at p can be compensated by the out-degrees, which are $\Omega(\Sigma_s)$ and unconstrained in parity. The even perfect numbers illustrate the escape route: their graph is the 2-cycle with $w(2,M_p) = 1$ and $w(M_p,2) = p$, symmetric in shape but not in weight. The handshake theorem is therefore a clean statement of the fact that *perfectness needs directedness.* Notice also what the proof does *not* use: no cyclotomic arithmetic, only (L1) and the parity of Euler's exponents.

6.6.8 The maximum principle and multiperfect numbers

The parity argument is special to Euler's form. A different undirected argument, the discrete maximum principle, works for every multiperfect number. For a k-perfect number N ($\sigma(N) = kN$, $k \geq 2$) we use the vertex set $\widehat{S}$ of all primes dividing kN, exponents $a_r = v_r(N)$ (possibly 0 for primes dividing only k), and weights $w(s,r) = v_r(\sigma(s^{a_s}))$, as in Section 2.2. Comparing r-adic valuations in $\prod_s \sigma(s^{a_s}) = kN$ gives the column sums

$$\sum_s w(s,r) = a_r + v_r(k).$$

Lemma 6.17. *For every prime R, $\displaystyle\prod_{p \leq R} \frac{p}{p-1} \leq R$, the product taken over primes.*

Proof. For $R = 2$ both sides equal 2. If $R < R'$ are consecutive primes and the inequality holds at R, then the product at R' is at most $R \cdot \frac{R'}{R'-1} \leq R'$, because $R \leq R' - 1$. □

(The inequality is very generous: by Mertens' theorem the product grows only like $e^{\gamma} \log R$.)

Theorem 6.18 (Maximum-principle obstruction)**.** *No k-perfect number with $k \geq 2$ has a symmetric weight matrix on $\widehat{S}$. In particular no perfect number, even or odd, does.*

Proof. Suppose W is symmetric, so row sums equal column sums. Because every prime factor of every $\sigma(s^{a_s})$ lies in $\widehat{S}$, the row sum at s is $\Omega(\sigma(s^{a_s}))$, and therefore

$$\Omega\big(\sigma(s^{a_s})\big) = a_s + v_s(k) \qquad (s \in \widehat{S}). \tag{$*$}$$

Let $R = \max \widehat{S}$. If $a_R = 0$ then $R \mid k$, and $(*)$ reads $0 = v_R(k) \geq 1$, which is absurd; so $R \mid N$. The prime factors of $\sigma(R^{a_R})$ all lie in $\widehat{S}$ and are different from R, hence are smaller than R. So by $(*)$

$$R^{a_R} < \sigma(R^{a_R}) < R^{\,\Omega(\sigma(R^{a_R}))} = R^{\,a_R + v_R(k)},$$

which forces $v_R(k) \geq 1$, i.e. $R \mid k$ and $k \geq R$. On the other hand, all primes of N are at most R and $\sigma(p^a)/p^a < p/(p-1)$, so Lemma 6.17 gives $k = h(N) < \prod_{p \leq R} p/(p-1) \leq R$. Contradiction. □

Why call this a maximum principle? When W is symmetric, $\Delta = \operatorname{diag}(\deg) - W$ is the ordinary Laplacian of the multigraph Γ, and

$$\Delta\ell(s) = \sum_r w(s,r)\,(\log s - \log r) = \deg(s) \log s - \log \sigma(s^{a_s}) = v_s(k) \log s - \eta_s,$$

where $\eta_s = \log h(s^{a_s}) > 0$. At the vertex R where the potential $\ell = \log$ is largest, every term $\log R - \log r$ is positive, so $\Delta\ell(R) > 0$. That positivity can only be paid for by the term $v_R(k) \log R$, which forces $R \mid k$; and the abundancy bound says k is far too small to be divisible by the largest prime. For the even perfect numbers the failure of symmetry is visible at a glance: $w(2, M_p) = 1$ but $w(M_p, 2) = p$.

6.6.9 The undirected Poisson problem without arithmetic

The last two theorems use the arithmetic of W. What if we keep only the *shape* of the Poisson equation of Section 3.3, $L\ell = -\delta$ with δ positive away from one vertex, but allow an arbitrary undirected graph? Then the problem becomes easy in the other direction: such graphs always exist, and the maximum principle tells us exactly where the sink must sit.

Proposition 6.19. *Let S be a finite set of primes with $|S| \geq 2$ and let $p \in S$. There is a connected undirected graph on S with positive integer edge weights whose Laplacian satisfies*

$$\Delta\ell(q) < 0 \quad (q \neq p), \qquad \Delta\ell(p) > 0, \qquad \ell_s = \log s,$$

if and only if $p = \max S$.

Proof. If q is the largest prime, then $\Delta\ell(q) = \sum_r w(q,r)\log(q/r) > 0$ because q has at least one neighbor and every neighbor is smaller; so the largest prime must be p. Conversely, if $p = \max S$, the star with center p and unit weights works: $\Delta\ell(q) = \log(q/p) < 0$ for each leaf q, and $\Delta\ell(p) = \sum_{q\neq p}\log(p/q) > 0$. □

Example 6.20. On the path 3—7—5 with $p = 7$,

$$\Delta\ell(3) = \log\tfrac{3}{7} < 0, \qquad \Delta\ell(5) = \log\tfrac{5}{7} < 0, \qquad \Delta\ell(7) = \log\tfrac{49}{15} > 0.$$

So the analogue of the OPN conjecture for bare undirected graphs is *false*: the Poisson sign pattern is realized on every prime set whose largest element is the sink. This is a useful calibration. Any proof of nonexistence must use the arithmetic that defines W, not just the shape of the Poisson equation. Note also that the directed operator $D - W$ has no maximum principle, so Proposition 6.19 says nothing about whether the Euler prime of an actual OPN is its largest prime.

6.6.10 The quadratic-form shadow of the directed equation

Finally, one can ask what part of the genuine directed equation $L\ell = -\delta$ survives symmetrization. A matrix and its symmetric part $\frac{1}{2}(L + L^{\mathsf{T}})$ have the same quadratic form, so the honest undirected shadow is the single number $\ell^{\mathsf{T}} L\ell$. Writing $\ell_s\ell_r = \frac{1}{2}\big(\ell_s^2 + \ell_r^2 - (\ell_s - \ell_r)^2\big)$ and using (L1) gives, for every OPN,

$$\boxed{\sum_{s,r} w(s,r)\log^2\frac{s}{r} \;+\; \sum_s \big(a_s - \Omega(\Sigma_s)\big)\log^2 s \;=\; -2\sum_s \delta_s \log s.}$$

The first term is a Dirichlet energy of the symmetrized multigraph and is nonnegative. The second is a "potential" term $V_s = a_s - \Omega(\Sigma_s)$ (in-degree minus out-degree), whose entries sum to 0 because both $\sum_s a_s$ and $\sum_s \Omega(\Sigma_s)$ equal $\Omega(N)$. Operators of the form "Laplacian plus potential" are discrete Schrödinger operators, so the identity says that the symmetric shadow of the OPN equation is a Schrödinger rather than a pure Laplacian identity.

Example 6.21 (The identity for $N = 120$)**.** The same computation for a k-perfect number gives the potential $V_s = a_s - \Omega(\sigma(s^{a_s})) - v_s(k)$ and right side $-2\sum_s \eta_s \log s$. For $120 = 2^3\cdot 3\cdot 5$ (Example 2.12) the potentials are $V = (1, -2, -1)$ at $(2, 3, 5)$, the energy term is about 2.433, the potential term about -4.524, and both sides equal about -2.090. (The identity was also checked numerically on the fourteen smallest multiperfect numbers.)

In the σ-symmetric case $V \equiv 0$, and the identity would say that $\sum_{q\neq p}\delta_q\log(p/q)$ is a positive energy, consistent with Proposition 6.19. For a genuine OPN, however, V has entries of both signs, so this one scalar identity cannot produce a contradiction on its own. It is a consequence of the directed equation, not an equivalent form of it.

6.6.11 Which undirected version is useful for which question?

It is helpful to keep the hierarchy straight:

Undirected object	What it remembers	Exact?
Underlying simple graph U_N	Which primes interact at all	No
Source–target bipartite graph B_N	Every entry $w(s,r)$ of W	Yes
Obligation–prime bipartite graph	Primitive-divisor eligibility and Hall capacity	Exact for the Hall relaxation
Antisymmetric current J	Logarithmic divergence and all cut identities	Exact for that logarithmic data
Symmetric block matrix $\mathcal{A}_W$	The full matrix W and its singular values	Yes
Symmetric dilation $\mathcal{L}_{\mathrm{sym}}$	The full Poisson operator L	Yes
σ-symmetric multigraph Γ	Everything, but only when $W = W^{\mathsf{T}}$	Yes, but never perfect
Bare undirected Poisson problem	Sign pattern of $\Delta\ell$ only	No (always solvable)
Quadratic form $\ell^{\mathsf{T}} L \ell$	One scalar identity	No

The main lesson is that an undirected reformulation need not discard the arithmetic asymmetry. The source–target construction transfers that asymmetry into the bipartition, while the current formulation transfers it into the sign of an antisymmetric edge variable. These are mathematically canonical ways to bring matching theory, ordinary cut theory, singular values, and signed-network ideas into the OPN problem without pretending that the original arrows were irrelevant.

6.7 Divisor-graph reformulations

This subsection asks which reformulations of the OPN conjecture are available in the language of divisor graphs (Section 2.7). The short answer: unlabeled graphs carry no information about perfectness, exact reformulations appear as soon as the vertices keep their labels, and the σ-graph is what remains of the labeled Hasse graph once its product structure is factored out.

6.7.1 Unlabeled divisor graphs cannot detect perfectness

Section 2.7 showed that the unlabeled graphs $D(n)$ and $H(n)$ depend only on the exponents of n (Example 2.29). For odd numbers this has a sharp consequence.

Proposition 6.22. *For every odd $n > 1$ there is an odd deficient n' with $D(n') \cong D(n)$ and $H(n') \cong H(n)$. Hence any property of unlabeled graphs that holds for $D(N)$ (or $H(N)$) of an odd perfect number N also holds for the divisor graph of some deficient odd number, so no such property can characterize odd perfect numbers.*

Proof. If n has k distinct prime factors, replace them by k distinct primes $q_1, \dots, q_k > 2k$, keeping the exponents. The new number n' has the same unlabeled divisor graphs, and

$$h(n') < \prod_{i=1}^{k} \frac{q_i}{q_i - 1} \le \left(1 + \frac{1}{2k}\right)^k < e^{1/2} < 2. \qquad \square$$

6.7.2 Exact reformulations with labeled vertices

Once the vertices keep their integer labels, exact reformulations are easy to write down. All of them rest on the identity $h(n) = \sigma_{-1}(n) = \sum_{d|n} 1/d$ (Proposition 1.6).

(A) **Label sum.** n is perfect if and only if the vertex labels of $D(n)$ sum to twice the largest label.

(B) **Reciprocal labels.** n is perfect if and only if $\sum_{d \in V(D(n))} 1/d = 2$. For 6 this is $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{6} = 2$; for 28 it is $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{7} + \frac{1}{14} + \frac{1}{28} = 2$; for the odd abundant number 945 of Example 2.11 the sum is 128/63.

(C) **Partition function of the labeled Hasse graph.** With the vertex weights m of Section 2.7, $Z(n) = \sigma(n)$. So n is perfect if and only if $Z(n) = 2\,m(n)$. For the grid of Example 2.28, $Z(28) = 56 = 2 \cdot 28$.

(D) **An unweighted encoding.** Form a bipartite graph with one vertex for each divisor d on one side and the vertices $1, \ldots, n$ on the other, joining d to the first d of them. Then $\deg d = d$, the number of edges is $\sigma(n)$, and n is perfect if and only if the graph has exactly $2n$ edges.

In each case the OPN conjecture becomes the statement that no odd $n > 1$ satisfies the condition. Formulation (C) is the most informative: it says that *the OPN conjecture is equivalent to the assertion that no Cartesian product of paths $P_{a_1+1} \square \cdots \square P_{a_k+1}$, with distinct odd primes assigned to its coordinate directions, has partition function equal to twice the weight of its top vertex.* It separates the shape of the Hasse graph, which Proposition 6.22 shows is irrelevant on its own, from the prime labels, where all the arithmetic lives. Formulation (D) is deliberately artificial. It shows that *some* graph-theoretic reformulation is always available, so the real question is never whether a reformulation exists, but whether the graph structure does any work.

6.7.3 The Hasse graph and the σ-graph together

Section 2.7 explained how the labeled Hasse graph $H(n)$ and the σ-graph divide the work: the Hasse graph records that $\sigma(n)$ is a product of chain sums, and the σ-graph records how each chain sum factors into primes. Perfectness is the condition that, after the chain sums are factored and multiplied together, every prime returns to the top vertex n with exactly its original exponent, plus one extra factor of 2. In the language of the labeled Hasse graph this is $Z(n) = 2m(n)$; in the σ-graph language it is $d(n) = e_2$ (Corollary 3.7). Example 2.33 shows both for $n = 28$.

6.7.4 Summary of the divisor-graph reformulations

Graph	Exact?	What carries the information
Unlabeled $D(n)$ or $H(n)$	No	Only the exponent pattern (Proposition 6.22)
Labeled $D(n)$, sums (A), (B)	Yes	A restatement of $\sigma(n) = 2n$
Labeled Hasse graph $H(n)$, partition function (C)	Yes	Prime labels on a product of paths
Bipartite encoding (D)	Yes	An edge count, by construction
Extended σ-graph, defect vector	Yes	The prime factorization of each chain sum

The lesson matches the rest of these notes. Graph shape alone carries no information about perfectness, and every exact reformulation puts the arithmetic into labels or weights. Among the

labeled structures, the σ-graph is the most economical: it discards the product structure, which is the same for every integer, and keeps only the prime factorizations of the chain sums, which is where any obstruction to an odd perfect number would have to be found.

7 Summary: odd perfect numbers, Poisson equations, and harmonic morphisms

7.1 Overview

This final section collects the main conclusions of the notes and develops the one reformulation that connects the OPN problem with the theory of harmonic morphisms and equitable partitions of graphs [16, 17, 18]. The guiding question is whether the OPN conjecture can be expressed in the language of edge-weighted graphs, equitable partitions, and harmonic morphisms. The answer is "partly," and the part that fails is instructive.

The main conclusions of the preceding sections are the following.

- **The σ-graph is a faithful encoding.** Valuation balance (Lemma 4.2) together with σ-closure is the equation $\sigma(N) = 2N$ written one prime at a time (Section 4).
- **Edges come in three arithmetic kinds.** Lemma 4.3 classifies them as order-one, Wieferich-type, and multiplier contributions. The order-one case is missing from Theorem 5.1 of the preprint [27] of Rodgers, and Example 4.1 shows that it occurs.
- **The proposed lemmas are correct but mostly classical.** The cut identity and the orientation toward the Euler prime are abundancy facts in graph language, and the two "breakthrough" claims are restatements of the conjecture rather than reductions of it (Section 5).
- **A few structural facts can be proved.** The σ-graph of an OPN is strongly connected and its Laplacian has rank $\omega(N)-1$ (Proposition 4.11, Corollary 4.13); most of the logarithmic mass of N circulates around cycles (Proposition 6.1); and no odd perfect or multiperfect number has a symmetric σ-graph (Theorems 6.16 and 6.18).
- **The construction is general.** Every integer $n > 1$ has an extended σ-graph whose defect vector encodes $h(n)$, and the OPN conjecture says that no odd n has defect vector e_2 (Corollary 3.7). Unlabeled divisor graphs, by contrast, cannot detect perfectness at all (Proposition 6.22).
- **The kernel of the Poisson operator is arithmetical.** An OPN σ-graph carries a canonical out-arithmetical structure (R_N, a) in the sense of Lorenzini, with a finite critical group $K(N)$, and R_N must be constant on the cells of any equitable quotient with constant exponents (Theorem 6.3, Proposition 6.7).
- **A realistic program targets structural inequalities.** Discharging and LP duality on vertex types can be benchmarked against the current record bounds on $\Omega(N)/\omega(N)$ (Section 6).

The rest of this section examines what quotient maps can and cannot do with the Poisson equation of Proposition 3.8.

7.2 What harmonic morphisms would do

Why try harmonic morphisms at all? In graph theory, a good quotient can replace a large graph by a smaller one while preserving enough of the Laplacian to transport harmonic functions and divisors. If the many prime vertices of an OPN were forced into arithmetic classes with identical local behavior, such a quotient could conceivably compress the Poisson equation to a much smaller obstruction. The difficulty is that multiplicative order is highly sensitive to the individual primes, so there is no obvious source of the symmetry needed for such a quotient.

Recall harmonic morphisms and in-equitable partitions from Section 2.4. For the σ-graph a quotient is a map $\varphi\colon S \to V(H)$ given by a partition of S into cells.

There are two ways one might try to transport the Poisson equation.

Pullback fails outright. Identity (B2) concerns potentials of the form $f \circ \varphi$, i.e. constant on fibers. But $\ell_s = \log s$ takes distinct values at distinct primes, so ℓ is never pulled back from a proper quotient.

Pushforward works, and produces something spoof-like.

Proposition 7.1. *Suppose the partition is in-equitable for W and a_s is constant ($= a_C$) on each cell C. Put $\Lambda_C = \sum_{r\in C} \log r = \log \Pi_C$, where $\Pi_C = \prod_{r\in C} r$. Then*

$$(a_C - b_{CC})\Lambda_C - \sum_{C'\neq C} b_{CC'}\Lambda_{C'} = -\sum_{s\in C} \delta_s,$$

i.e. $L_H\Lambda = -\varphi_*\delta$ with $L_H = \mathrm{diag}(a_C) - B$, and L_H again has zero column sums.

Proof. Sum row s of $L\ell = -\delta$ over $s \in C$: $\sum_{s\in C} a_s\ell_s = a_C\Lambda_C$, and $\sum_{s\in C}\sum_r w(s,r)\ell_r = \sum_{C'}\sum_{r\in C'}\ell_r\sum_{s\in C} w(s,r) = \sum_{C'} b_{CC'}\Lambda_{C'}$. Column sums: $\sum_C b_{CC'} = \sum_s w(s,r) = a_r = a_{C'}$. □

Remark 7.2 (Left-equitability and the Descartes warning). In standard algebraic graph theory terminology, a partition where the column sum $\sum_{s\in C} w(s,r) = b_{CC'}$ is independent of $r \in C'$ is a *left-equitable* (or column-equitable) partition for W: the characteristic vectors $\mathbf{1}_C$ span a W^{T}-invariant subspace. Proposition 7.1 is therefore an ordinary equitable-quotient calculation adapted to the in-degree convention.

The Descartes spoof should *not* be described as a literal equitable quotient of a genuine six-prime valuation graph on $\{3, 7, 11, 13, 19, 61\}$. The incoming factors $19^2\cdot 61$ do combine to 22021, but after splitting the pseudo-prime the outgoing row would be governed by $\sigma(19^2)\sigma(61)$, whereas the spoof uses $(22021+1)/2$. Example 5.1 shows the correct statement: one can collapse the incoming prime factors into a *logarithmic* coordinate using real coefficients θ and ψ. That operation preserves the displayed linear Poisson balance but is not an equitable quotient of an integer valuation matrix. The failure is itself useful: it pinpoints exactly where primality enters the local divisor-sum dynamics.

Exponentiating Proposition 7.1, $\prod_{s\in C}\Sigma_s = \prod_{C'}\Pi_{C'}^{b_{CC'}}$. Thus a genuine equitable quotient satisfies the same kind of multiplicative *balance* equations as a network of composite pseudo-primes Π_C. It need not, however, satisfy the stronger local identity $\Sigma_C = \sigma(\Pi_C^{a_C})$; that extra divisor-sum requirement is precisely where primality and the individual components matter. The right word is therefore *spoof-like*, not "exactly a spoof." Quotienting forgets multiplicative orders, Zsigmondy layers, and LTE at the individual primes, which are exactly the kinds of facts that distinguish real OPNs from composite pseudo-prime constructions.

Example 2.12 shows that even small σ-graphs need not admit equitable partitions (Section 2.4).

In-equitable partitions push the Poisson equation *forward*. The transposed notion, out-equitable partitions, interacts instead with the kernel of L: by Proposition 6.7, the canonical out-arithmetical structure R_N of Theorem 6.3 must be constant on the cells of any out-equitable partition with constant exponents. This is the directed counterpart of the pullback of arithmetical structures along harmonic morphisms in [18], and it turns the uniqueness of R_N into a practical test for equitability.

7.3 Limitations of the quotient approach

Three obstacles stand in the way of a proof along these lines.

1. **No source of symmetry.** Edge existence depends on whether $\mathrm{ord}_r(s) \mid a_s + 1$ (Lemma 4.3). That is not constant on any natural cells, so equitability would have to be *proved*, which is already hard arithmetic.

2. **Over $\mathbb{R}$ the linear equation is empty.** For the ordinary Laplacian of a connected undirected graph, every degree-zero real divisor lies in the image. For a general directed operator this would need an extra rank hypothesis, but for an OPN that hypothesis holds automatically: G_N is strongly connected and $\operatorname{rank} L = \omega(N) - 1$ (Corollary 4.13), so *every* real vector of degree zero is L of something. Proposition 6.19 makes the same point for undirected graphs: the bare sign pattern of the Poisson equation is always realizable. The content therefore lies entirely in W being an integer factorization matrix of cyclotomic values and in ℓ being logarithms of actual primes. Harmonic-morphism functoriality preserves linear structure but is largely blind to that arithmetic content.

3. **Directed and asymmetric.** The harmonic-morphism theory in Baker–Norine [2], Urakawa [29], and Joyner–Melles [16] and Melles–Joyner [17, 18] is for undirected graphs. One would need a weighted-digraph theory that interacts well with a global sink. Directed chip-firing with a sink [15] fits the unique-Euler-sink picture of Lemma 4.9, but works with integer firing vectors, and $\log s$ is not an integer.

7.4 Two directions that remain open

The cyclotomic blow-up as a bipartite factor digraph. The cyclotomic decomposition of Remark 1.27 becomes graph-theoretic through a bipartite subdivision digraph $\widetilde{G}$ on the vertex set $S \sqcup \mathcal{O}$:

- For each prime $q \in Q$ and each divisor $d \mid e_q$ ($d > 1$), place a directed edge $q \to (q, d)$ of weight 1 (or logarithmic weight $\log \Phi_d(q)$).
- From each obligation vertex $(q, d) \in \mathcal{O}$ to each prime $r \in S$, place a directed edge $(q, d) \to r$ of weight $\widetilde{w}((q, d), r) = v_r(\Phi_d(q))$. (Treat the Euler vertex p analogously via the odd part of $\Phi_d(p)$ for $d \mid \alpha + 1$.)

By Remark 1.27, summing the 2-step paths $q \to (q, d) \to r$ over all $d \mid e_q$ recovers the exact edge weight $w(q, r)$ of G_N, so G_N is the 2-step path condensation of $\widetilde{G}$. Moreover, collapsing $(q, d) \mapsto q$ gives a projection $\widetilde{G} \to G_N$ whose fibers have size $\sigma_0(e_q) - 1$; the Zsigmondy count $\sigma_0(e_q) \leq \omega(N)$ reads like a fiber-counting statement, and the Hall deficiency condition (H) of Claim A is simply Hall's condition on the primitive-prime subgraph of the bipartite half $\mathcal{O} \to S$. This is the one map that arises canonically from the arithmetic rather than by arbitrary partitioning. Whether it satisfies a useful weighted harmonic property remains open.

Example 7.3 (Two blow-up vertices)**.** For 13^2 one has $e_{13} = 3$ and

$$\Phi_3(13) = 183 = 3 \cdot 61.$$

The blow-up replaces the direct edges out of 13 by

$$13 \longrightarrow (13, 3) \longrightarrow \{3, 61\}.$$

The edge to 61 is the primitive/order-3 part, while the edge to 3 is the exceptional order-one cyclotomic contribution discussed in Lemma 4.3. Thus the intermediate vertex remembers that two arithmetically different kinds of factors arose from the *same* cyclotomic layer.

For 3^4, by contrast,

$$\Phi_5(3) = 121 = 11^2,$$

so the blow-up contains

$$3 \longrightarrow (3, 5) \xrightarrow{2} 11.$$

This second example shows why retaining the weight on the obligation-to-prime edge matters: one cyclotomic layer can consume two units of the target's valuation capacity.

Riemann–Hurwitz and the cycle space. For a harmonic morphism, the Riemann–Hurwitz formula relates the genera of source and target [2, 17]. The genus (first Betti number) of G_N is the dimension of its cycle space, which is where the circulation part of the flow decomposition in Section 6 lives. Proposition 6.1 shows that more than two thirds of the logarithmic mass of N, and asymptotically almost all of it, travels around directed cycles, so the cycle space is where most of the structure is. Genus bounds relating $\widetilde{G}$ and G_N could nevertheless limit how the cyclic part is organized, if a suitable harmonic map could be found. In plain language, the hope would be to prove that the many cyclotomic layers force more independent cycles upstairs than a low-complexity quotient can support. At present this is only a structural analogy: graph genus counts independent cycles, but it does not remember whether those cycles satisfy the multiplicative-order and valuation restrictions that make an OPN graph arithmetically special.

7.5 The undirected case

Since harmonic morphisms are best understood for undirected graphs, a natural question is whether an undirected version of the σ-graph can exist. The answer is sharp and fits in a paragraph. Call N *σ-symmetric* if $w(s, r) = w(r, s)$ for all primes $s, r \mid N$, so that W is the adjacency matrix of an ordinary undirected multigraph. Such local symmetry does occur: $\sigma(3^2) = 13$ and $\sigma(13^2) = 3 \cdot 61$, so 3 and 13 supply each other exactly once. But no odd perfect number can be σ-symmetric, and the proof is the handshake lemma. By valuation balance, the degree of the vertex s would be its exponent a_s, and Euler's form says that exactly one exponent, the special one, is odd. An undirected graph cannot have exactly one vertex of odd degree.

A maximum-principle argument extends this to every multiperfect number (Theorem 6.18): at the largest prime the Laplacian of the potential log is forced to be positive, and the only thing that can pay for that positivity is a factor of the largest prime in the abundancy k, which is far too small. On the other hand, once the arithmetic is dropped and only the sign pattern of the Poisson equation is kept, undirected examples exist on every prime set whose largest element is the sink (Proposition 6.19). Both facts point the same way: the directedness of the σ-graph is not a technical nuisance. It is where the arithmetic lives.

7.6 Conclusions

An odd perfect number is a weighted digraph on which log *is a potential for a degree-zero divisor that is positive away from one vertex.*

The reformulation is natural, and equitable quotients of it produce spoof-like multiplicative balance systems, although they need not preserve the local divisor-sum identity of a genuine spoof. This is a precise way of saying why linear-algebraic graph theory, on its own, cannot see the arithmetic that distinguishes primes from composite pseudo-primes. The undirected version, by contrast, is too rigid to be interesting: a symmetric σ-graph is ruled out by the handshake lemma. Whether some arithmetic partition is *forced* to be equitable remains open; the absence of any natural source of symmetry suggests that the answer is negative.

A Reader's glossary of technical terms

The notes use terminology from several subjects at once. This appendix is a reference glossary for readers who know one of number theory, graph theory, or optimization better than the others. Definitions already given above are repeated only when doing so prevents ambiguity. None of the definitions in this section asserts the existence of an OPN.

Number-theoretic terms

Divisibility and exact divisibility.
The notation $a \mid b$ means that $b = ac$ for some integer c. The notation $q^a \parallel N$ means $q^a \mid N$ but $q^{a+1} \nmid N$; thus $a = v_q(N)$.

Greatest common divisor and coprime integers.
The greatest common divisor $\gcd(a, b)$ is the largest positive integer dividing both a and b. Integers a and b are *coprime* when $\gcd(a, b) = 1$.

Congruence and residue class.
The notation $a \equiv b \pmod{m}$ means $m \mid (a - b)$. Two integers congruent modulo m represent the same residue class. Statements such as $q \equiv 1 \pmod 3$ therefore classify primes by their remainder modulo a small integer.

r-adic valuation.
For a prime r and a nonzero integer x, $v_r(x)$ is the largest integer $e \geq 0$ such that $r^e \mid x$. It records the multiplicity of r in the prime factorization of x.

Multiplicative order.
If r is prime and $r \nmid s$, then $\mathrm{ord}_r(s)$ is the least positive integer d such that $s^d \equiv 1 \pmod r$. It necessarily divides $r - 1$.

Prime support, ω, and Ω.
The *prime support* of an integer is the set of primes dividing it. The function $\omega(N)$ counts distinct prime divisors, while $\Omega(N)$ counts prime factors with multiplicity. Thus if $N = \prod_i p_i^{a_i}$, then $\omega(N)$ is the number of indices i and $\Omega(N) = \sum_i a_i$.

Divisor functions σ_r, σ_0, σ.
$\sigma_r(n) = \sum_{d \mid n} d^r$. The value $\sigma_0(n)$ (often written $\sigma_0(n)$) is the number of positive divisors of n, and $\sigma = \sigma_1$ is their sum. If $n = \prod_i p_i^{a_i}$, then $\sigma_0(n) = \prod_i (a_i + 1)$.

Euler totient function φ.

The value $\varphi(n)$ counts the integers $1 \le j \le n$ with $\gcd(j, n) = 1$. Equivalently, it is the order of the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$. The degree of $\Phi_n(x)$ is $\varphi(n)$, which explains the size heuristic $\Phi_n(q) \approx q^{\varphi(n)}$ for large q.

Cyclotomic polynomial.

The dth cyclotomic polynomial $\Phi_d(x)$ is the monic (leading coefficient 1) polynomial whose complex roots are the primitive dth roots of unity, meaning numbers z with $z^d = 1$ but $z^j \neq 1$ for $1 \le j < d$. It is characterized by $x^n - 1 = \prod_{d|n} \Phi_d(x)$. In these notes the factorization of $\sigma(q^a) = (q^{a+1} - 1)/(q - 1)$ into cyclotomic factors is used to locate prime divisors of local divisor sums.

Primitive prime divisor.

A prime r is a *primitive prime divisor* of $s^d - 1$ if $r \mid s^d - 1$ but $r \nmid s^j - 1$ for every $1 \le j < d$. Equivalently, $\mathrm{ord}_r(s) = d$. "Primitive" is relative to the fixed base s and exponent d; it does *not* mean that r is a new prime globally in the σ-graph.

Abundancy.

The abundancy of n is $h(n) = \sigma(n)/n$. A number is perfect precisely when $h(n) = 2$. Because σ is multiplicative on coprime arguments, h is also multiplicative on coprime arguments.

Multiplicative arithmetic function.

An arithmetic function f is multiplicative if $f(ab) = f(a)f(b)$ whenever $\gcd(a, b) = 1$. This is weaker than being *completely* multiplicative, which would require the identity for all a, b. The divisor-sum function σ and abundancy h are multiplicative, not completely multiplicative.

Component and special component.

A prime power $q^a \parallel N$ is called a *component* of N. In Euler's form for an OPN, p^α is the unique *special* (or Euler) component with $p \equiv \alpha \equiv 1 \pmod 4$; all other prime-power components have even exponent and are called *nonspecial*.

Unitary divisor.

A divisor $M \mid N$ is unitary if $\gcd(M, N/M) = 1$. For $N = \prod_i p_i^{a_i}$, a unitary divisor is obtained by choosing some of the entire prime-power components $p_i^{a_i}$ and omitting the rest.

σ-closure.

A set S of odd primes with assigned exponents is *σ-closed* if every odd prime divisor of every relevant local divisor sum $\sigma(s^{a_s})$ also lies in S. For an OPN, after the unique factor 2 is removed from the Euler component, σ-closure follows from $\sigma(N) = 2N$.

σ-chain or factor chain.

A sequence $q_0, q_1, \ldots$ with $q_{i+1} \mid \sigma(q_i^{a_{q_i}})$ at each step is a σ-chain. In G_N it is simply a directed walk. Factor-chain arguments use such successive divisibility to force additional prime components.

$\sigma_{m,n}$-pair.

Following Bibby–Vyncke–Zelinsky, primes p, q form a $\sigma_{m,n}$-pair when $q \mid \sigma(p^m)$ and $p \mid \sigma(q^n)$; in the present arrow convention this is a directed 2-cycle when the exponents at p and q are m and n.

Lifting the exponent (LTE).
LTE is a family of valuation identities that converts divisibility of $x - y$ into an exact formula for $v_r(x^n - y^n)$ under suitable hypotheses. Only the odd-prime form stated in Section 1.5 is needed here.

Wieferich-type lifting.
The phrase is used informally here for the exceptional situation $v_r(s^d - 1) > 1$ when $d = \mathrm{ord}_r(s)$. It means that the base cyclotomic layer already contributes more than one copy of r, before any extra multiplier $v_r(n/d)$ is counted. It is "Wieferich-type" terminology, not a claim that the classical base-2 Wieferich-prime condition is present.

Multiplier contribution.
When $d = \mathrm{ord}_r(s) > 1$ divides $n = a_s + 1$, LTE gives a term $v_r(n/d)$ in the edge weight. This extra valuation is called the multiplier contribution; it is additional weight on an existing order-d edge, not a new kind of edge.

Nagell–Ljunggren equation.
This is the Diophantine equation

$$\frac{x^n - 1}{x - 1} = y^m, \qquad x, y > 1, \quad n > 2, \quad m > 1,$$

which asks when a repunit-like cyclotomic product is a perfect power. The general equation is not completely solved; only special cases and strong restrictions are known.

S-unit and S-unit equation.
Given a finite set of primes $\mathcal{S}$, an $\mathcal{S}$-unit is a nonzero rational number whose numerator and denominator have no prime factors outside $\mathcal{S}$. An $\mathcal{S}$-unit equation is a Diophantine equation, such as $x + y = 1$, in which the unknowns are constrained to be $\mathcal{S}$-units. In this paper the phrase indicates that a cyclotomic value is forced to use only a prescribed finite prime support.

k-perfect or multiperfect number.
A positive integer N is k-perfect if $\sigma(N) = kN$. The usual perfect numbers are the case $k = 2$; "multiperfect" is a common collective term for integers satisfying such an equation for an integer $k \geq 2$.

Mersenne prime.
A Mersenne prime is a prime of the form $M_p = 2^p - 1$. Euclid–Euler theory says every even perfect number is $2^{p-1}M_p$ for such a prime M_p.

Spoof odd perfect number; spoof prime.
In these notes, a *spoof odd perfect number* means a Descartes-type formal perfect factorization in which a composite odd factor is deliberately treated as though it were prime when the local divisor-sum formula is applied. In the common one-spoof-factor formulation, an odd integer $n = km$ satisfies

$$\sigma(k)(m + 1) = 2km,$$

with the composite m playing the role of a *spoof prime*, also called a pseudo-prime or quasi-prime factor. This identity does not imply that the ordinary divisor sum $\sigma(km)$ equals $2km$, because m is not actually prime. Descartes' $m = 22021 = 19^2 \cdot 61$ is the standard example; see Dittmer [13] and the BYU Computational Number Theory Group [9].

Diophantine equation.
A Diophantine equation is a polynomial or arithmetic equation in which the unknowns are required to be integers (or, in a specified variant, rational numbers). The adjective emphasizes that integrality and prime factorization, not merely real-valued solvability, are essential.

Graph-theoretic and linear-algebraic terms

Directed graph (digraph), vertex, and edge.
A directed graph consists of vertices together with ordered pairs $s \to r$ called directed edges or arcs. The first endpoint s is the *source* and the second r the *target*. A weighted digraph assigns a numerical weight to each edge.

Loop, walk, path, and directed cycle.
A loop is an edge from a vertex to itself. A directed walk follows arrows and may repeat vertices. A directed path is a walk with no repeated vertices (apart from the usual allowance when closing a cycle). A directed cycle is a closed directed path.

In-degree and out-degree.
The weighted in-degree of a vertex is the sum of the weights of edges entering it; the weighted out-degree is the corresponding sum over edges leaving it. For G_N, valuation balance fixes the in-degree of r to be a_r, but does not similarly fix its out-degree.

Strong connectivity and strongly connected component (SCC).
Vertices u and v are strongly connected if each is reachable from the other by a directed path. A strongly connected component is a maximal set of pairwise strongly connected vertices.

Arithmetical structure and arithmetical critical group.
Following Lorenzini, an arithmetical structure on a connected graph is a pair of positive integer vectors (R, S) with $\gcd R = 1$ and $(\operatorname{diag}(S) - A)R = 0$. The matrix $\operatorname{diag}(S) - A$ is its generalized Laplacian, and the torsion of its cokernel is the arithmetical critical group. The natural structure is $R \equiv 1$, $S = \deg$. Section 6.5 extends the notion to weighted digraphs.

Handshake lemma.
In any finite undirected (multi)graph the sum of the vertex degrees is twice the number of edges. Consequently the number of vertices of odd degree is even. In these notes this elementary fact rules out a symmetric σ-graph for an OPN, whose Euler form has exactly one odd exponent.

Irreducible matrix and harmonic function of a Markov chain.
A nonnegative square matrix is irreducible when its support digraph is strongly connected. A function y is harmonic for a stochastic matrix M when $y = My$, i.e. each value is the weighted average of the values at the next step. For an irreducible chain the only harmonic functions are the constants.

Maximum principle.
For a graph Laplacian $\Delta f(x) = \sum_y w(x, y)(f(x) - f(y))$ with nonnegative weights, $\Delta f(x) \geq 0$ at any vertex where f attains its maximum, with strict inequality if some neighbor has a smaller value. This discrete maximum principle has no general analogue for the nonsymmetric operator $D - W$ of these notes.

Condensation digraph.
The condensation of a digraph is obtained by collapsing each SCC to one vertex. The condensation is always a directed acyclic graph (DAG), meaning that it contains no directed cycle.

Source and sink components.
A source component of the condensation has no incoming edge from another component; a sink component has no outgoing edge to another component. This is distinct from the flow-theoretic use of "source" and "sink," where a vertex has positive or negative divergence.

In-arborescence.
An in-arborescence rooted at p is a directed spanning tree in which every other vertex has a unique directed path toward p. Thus all tree arrows point, globally, into the root. The directed Matrix–Tree theorem counts weighted rooted arborescences by Laplacian cofactors.

Undirected graph, weighted graph, and multigraph.
An *undirected graph* has edges $\{u, v\}$ with no preferred direction. A weighted undirected graph assigns a numerical weight to each edge. A *multigraph* allows several parallel edges between the same two vertices. Thus an integer edge of weight m can often be pictured as m parallel unit-weight edges. When we later "forget the arrows" of G_N, we obtain an undirected support graph, but that operation loses information and is not an equivalent reformulation.

Bipartite graph, biadjacency matrix, and subdivision.
A *bipartite graph* has its vertices separated into two color classes U and V, with edges only between the classes. Its *biadjacency matrix* B has one row for each $u \in U$, one column for each $v \in V$, and entry B_{uv} equal to the edge weight between them. Unlike an ordinary adjacency matrix, a biadjacency matrix need not be symmetric. This is exactly what lets a bipartite graph remember a directed matrix. A *subdivision* inserts new vertices along edges. The cyclotomic blow-up $\widetilde{G}$ is bipartite between prime vertices and obligation vertices and can be viewed as subdividing each prime-to-prime factorization relation through its cyclotomic layer.

Source–target bipartite graph.
Given a digraph on S, make two labeled copies $S^- = \{s^- : s \in S\}$ and $S^+ = \{r^+ : r \in S\}$. Replace a directed edge $s \to r$ of weight m by an undirected bipartite edge s^-–r^+ of weight m. If the two color classes and the prime labels are retained, this construction is reversible: its biadjacency matrix is the original directed weight matrix.

Adjacency/weight matrix.
For a weighted digraph, the matrix $W = (w(s, r))$ records the weight from source s to target r. In these notes rows index sources and columns index targets; this convention matters when distinguishing row sums from column sums.

Support digraph.
The support digraph of a weighted matrix or weighted digraph forgets the numerical weights and retains exactly those directed edges for which the weight is nonzero. Thus the support of W has an edge $s \to r$ exactly when $w(s, r) > 0$.

Column-stochastic matrix.
A nonnegative matrix is column-stochastic if every column sums to 1. Equivalently, $\mathbf{1}^\mathsf{T} P = \mathbf{1}^\mathsf{T}$. Such a matrix has spectral radius 1 and eigenvalue 1.

Eigenvalue, eigenvector, nullspace, rank, and invariant subspace.
An eigenvector of a matrix A is a nonzero vector x satisfying $Ax = \lambda x$; λ is its eigenvalue. The nullspace consists of vectors sent to 0. A left-nullvector satisfies $y^\mathsf{T} A = 0$. The rank is the dimension of the image of A. A subspace is invariant under A if A maps it into itself.

Linear independence and injectivity.
Vectors or numbers x_i are linearly independent over a field (here usually $\mathbb{Q}$) if $\sum_i c_i x_i = 0$ with coefficients in that field forces all $c_i = 0$. A map is injective if two different inputs can never have the same output. The later logarithm remark uses unique factorization to show that distinct rational prime-exponent data remain distinct after taking logs.

Matrix minor and cofactor.
A minor of a matrix is the determinant of a square submatrix obtained by deleting selected rows and columns. A cofactor is a signed codimension-one minor. Matrix–Tree theorems identify particular Laplacian cofactors with weighted counts of spanning trees or directed arborescences.

Spectral radius, Perron eigenvalue, and spectral gap.
The spectral radius $\rho(A)$ is the largest absolute value of an eigenvalue of A. For a nonnegative stochastic matrix, 1 is the distinguished Perron eigenvalue. Informally, a spectral gap measures the separation between this leading eigenvalue and the remaining spectrum; the exact definition depends on the normalization and, for directed graphs, may involve singular values or a symmetrized operator.

Singular value, symmetric dilation, and symmetrization.
The singular values of A are the nonnegative square roots of the eigenvalues of $A^{\mathsf{T}}A$ (or A^*A over $\mathbb{C}$). They can measure expansion when a directed matrix is not symmetric. A particularly useful *symmetric dilation* is

$$\mathcal{S}(A) = \begin{pmatrix} 0 & A \\ A^{\mathsf{T}} & 0 \end{pmatrix}.$$

Its nonzero eigenvalues are the singular values of A together with their negatives. Thus a nonsymmetric arithmetic matrix can be embedded, without losing its entries, into a symmetric operator on a doubled vertex set.

Markov operator.
A Markov operator is a stochastic matrix acting on probability distributions (or, with the transpose convention, on functions). Its entries describe one-step transition probabilities of a Markov chain. In this manuscript WD^{-1} is column-stochastic, but its entries arise from normalized valuations rather than from an independently given random process.

Graph Laplacian and normalized Laplacian.
A graph Laplacian is a degree matrix minus an adjacency/weight matrix. Here $L = D_{\text{in}} - W$ has zero column sums. A normalized Laplacian rescales by degrees so that comparisons are insensitive to vertex degree; for directed graphs there are several inequivalent conventions, so the normalization must always be specified.

Cheeger/isoperimetric inequality.
A Cheeger-type inequality relates a graph's expansion across cuts to spectral data of an appropriate Laplacian or Markov operator. "Expansion" measures how difficult it is to separate a set of vertices from its complement by a small boundary. In directed graphs both the boundary notion and the normalization require extra choices.

Flow, circulation, divergence, and Kirchhoff law.
A flow assigns a nonnegative amount to directed edges. The divergence at a vertex is outflow minus in-flow. A circulation is a flow of zero divergence at every vertex. More generally an *antisymmetric current* on an undirected graph assigns a signed number $J_{uv} = -J_{vu}$ to each

oriented version of an undirected edge. Kirchhoff's current law says that the sum of the currents leaving a vertex equals the externally injected source strength there; in symbols, div $J = \delta$. The logarithmic edge weights in this paper form a near-circulation: nonspecial vertices have small positive divergence and the Euler vertex absorbs the total deficit.

Conductance and signed network.
In an ordinary electrical network an edge $\{u, v\}$ has nonnegative conductance c_{uv}, a potential V is assigned to vertices, and the current is $J_{uv} = c_{uv}(V_u - V_v)$. A *signed network* allows negative edge weights or conductances. The OPN current introduced later is an exact undirected current, but it need not come from nonnegative conductances with potential $V_s = \log s$; the primitive $13 \leftrightarrow 61$ cycle gives a concrete negative-conductance example.

Flow decomposition.
A finite nonnegative flow can be decomposed into flows supported on directed cycles together with flows along directed source-to-sink paths. The decomposition need not be unique.

Cut.
For $T \subseteq S$, the cut $(T, S \setminus T)$ separates T from its complement. In a directed weighted graph one distinguishes flow from T to $S \setminus T$ from flow in the reverse direction.

Potential and Poisson equation.
A potential is a function on vertices. An equation of the form $Lf = f_0$, where L is a Laplacian and f_0 is a prescribed source term, is a discrete Poisson equation. In the OPN reformulation the potential is the log-prime vector $\ell_s = \log s$ and the source term is $-\delta$.

Fredholm compatibility condition.
For a linear equation $Lx = b$ to be solvable, b must be orthogonal to every left-nullvector of L. This necessary condition is the finite-dimensional form of the Fredholm compatibility condition. If the left nullspace is spanned by $\mathbf{1}$, it reduces to the zero-sum condition $\mathbf{1}^{\mathsf{T}} b = 0$; for a general directed Laplacian additional conditions may be present.

Divisor on a graph and degree.
In graph divisor theory, a divisor is a formal integer linear combination of vertices, $D = \sum_v a_v(v)$, and its degree is $\deg D = \sum_v a_v$. When we speak informally of the real vector δ as a divisor, we have extended coefficients from $\mathbb{Z}$ to $\mathbb{R}$.

Principal divisor and chip-firing.
For an ordinary graph Laplacian Δ, a principal divisor is one of the form Δf for an integer-valued firing function f. Chip-firing is the combinatorial operation encoded by adding such Laplacian vectors. Our real-valued equation with potential $\log s$ is only a formal real analogue unless a compatible directed divisor theory is specified.

Harmonic morphism; horizontal and vertical edges.
A harmonic morphism of graphs is a map that has the same local horizontal multiplicity above every edge incident to the image vertex; the precise undirected definition used here is given later in Section 2.4. An edge is *horizontal* when it maps to an edge of the target and *vertical* when it collapses to a target vertex. A harmonic morphism is the graph analogue of a branched harmonic map and supports pullback/pushforward operations on divisors.

Partition, cell, fiber, and quotient.
A partition writes the vertex set as disjoint cells. Under a quotient map, the vertices in one

cell are identified; that cell is a fiber of the quotient. A quotient matrix records the induced interaction between cells when the necessary equitability conditions hold.

Equitable, in-equitable, and left-equitable partition.
A partition is equitable when vertices in the same cell see every other cell with the same prescribed aggregate weight. In this paper the relevant condition is equality of incoming column weights, called in-equitable or left-/column-equitable; the exact formula is given in Section 2.4.

Pullback and pushforward.
Given a map of graphs, pullback transports a function or divisor on the target back to the source, while pushforward aggregates source data on each target fiber. The later quotient calculation uses pushforward of the Poisson source term.

Cycle space, first Betti number, and graph genus.
For an undirected graph, the cycle space is the vector space generated by its cycles. Its dimension is the first Betti number $b_1 = |E| - |V| + c$, where c is the number of connected components. In the harmonic-morphism literature this number is often called the graph's genus. For a directed graph one usually applies this to the underlying undirected graph unless another convention is specified.

Riemann–Hurwitz formula for graphs.
This is the graph-theoretic analogue of the classical Riemann–Hurwitz formula for branched maps of Riemann surfaces: for a harmonic morphism it relates the genera of source and target graphs together with a ramification term. Here *ramification* measures the local failure of the map to be an unbranched, constant-degree covering.

Combinatorial and optimization terms

Matching, Hall condition, and Hall deficiency.
A matching in a bipartite graph is a set of edges no two of which share a vertex; it represents assigning distinct available targets to demand vertices. A *b-matching* or capacitated matching allows vertex v to be used up to a prescribed integer capacity $b(v)$. Thus a target may be used several times rather than only once. Hall's condition says that every set $\mathcal{A}$ of demand vertices has at least as much available capacity in its neighborhood $U(\mathcal{A})$ as the demand of $\mathcal{A}$. In the unit-capacity classical theorem this is $|U(\mathcal{A})| \geq |\mathcal{A}|$. A Hall deficiency is a set for which the required inequality fails. Claim A uses the capacitated form $|\mathcal{A}| \leq \sum_{r \in U(\mathcal{A})} a_r$.

Capacity.
Capacity is an upper bound on how much demand a vertex or edge is allowed to absorb. In the proposed cyclotomic matching model, a prime vertex r has valuation capacity a_r because its total incoming valuation is exactly a_r.

Discharging.
Discharging is a proof method in which one assigns an initial numerical charge to local objects, redistributes charge according to prescribed local rules, and proves that the final charges satisfy an inequality incompatible with the assumed global configuration. The total charge is conserved during the redistribution.

Linear program (LP).
A linear program optimizes a linear objective subject to linear equalities and inequalities. Here variables can encode numbers of vertex types and amounts of weight between types.

Relaxation.
A relaxation enlarges the feasible set by dropping difficult constraints, for example integrality, exact prime identities, or some nonlinear arithmetic conditions. If even the relaxation is infeasible, the original problem is infeasible; feasibility of the relaxation does not imply feasibility of the original arithmetic problem.

Primal problem, LP duality, and dual certificate.
The original linear program being optimized is called the *primal*; its associated optimization problem is the *dual*. Under standard hypotheses their optimal values agree. A feasible dual solution can therefore certify a bound for the primal problem; in a discharging interpretation, the dual variables often encode the coefficients of the charge-transfer rules. An *extremal primal solution* is a feasible primal assignment attaining, or lying at the boundary responsible for, the optimum.

Vertex type.
A vertex type is a deliberately coarse classification recording only selected local data—for example exponent class, residue classes modulo small primes, or whether the vertex is the largest prime. Type-based LPs forget the exact prime labels in order to obtain a finite optimization problem.

Reducible configuration.
A local configuration is called reducible when separate arithmetic reasoning shows that it cannot occur in a genuine OPN, or that it can be replaced by a configuration already controlled by the argument. This terminology mirrors the role of reducible configurations in discharging proofs such as the Four Color Theorem.

Feasibility and infeasibility.
A system of constraints is feasible if at least one assignment satisfies all of them and infeasible otherwise. A computational feasibility test of a relaxation is a sanity check, not a proof about configurations whose omitted arithmetic constraints were essential.

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